A Deep Learning Method for Field Boundary Delineation From Remote Sensing Imagery With High Boundary Connectivity

Yu Zhu, Yaozhong Pan, Tangao Hu, *Member, IEEE*, and Yao Liu

***Abstract*—Accurate delineation of agricultural field boundaries from remote sensing imagery is fundamental for precision agriculture and policymaking. However, existing deep learning methods struggle with this task due to the difficulty of balancing semantic context with fine detail. Their unique imaging features (e.g., intricate linear nature and frequent occlusions) further challenge the detection of continuous boundaries. To address these issues, we propose a high-connectivity and accuracy field boundary network (HCFNet). HCFNet employs a novel two-stage framework where a lightweight region segmentation task provides explicit semantic constraints to a dedicated boundary delineation task. The boundary task is powered by an enhanced edge network featuring four key innovations: a linearly connected deformable convolution (LCDConv) to capture fine boundary morphology, a nonlocal operation (NLO) to resolve local ambiguities using global context, a continuity restoration decoder (CRD) that expands the receptive field and bridge occluded segments, and a hierarchical attention fusion module (HAFM) for adaptive multiscale feature fusion. We conducted extensive experiments on diverse datasets, including high-resolution Gaofen-1 (GF-1), Google Earth, UAV imagery, and the global FTW benchmark. HCFNet consistently outperformed state-of-the-art models, achieving superior boundary accuracy ($F1$ score) and significantly improved connectivity. Our results demonstrate that HCFNet provides a robust and efficient solution for field boundary delineation, with its specialized modules proving particularly valuable in data-scarce scenarios common in practical applications. The code for HCFNet is available at https://github.com/BNU-zhu/HCFNet**

***Index Terms*—Agricultural remote sensing, automated delineation, edge network, field boundary, high spatial resolution image.**

Received 6 August 2025; revised 9 October 2025; accepted 27 October 2025. Date of publication 3 November 2025; date of current version 18 November 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFB3906200 and Grant 2023YFB3906201, in part by the National Natural Science Foundation of China under Grant 42192580 and Grant 42192581, and in part by the Open Fund of State Key Laboratory of Remote Sensing Science and Beijing Engineering Research Center for Global Land Remote Sensing Products under Grant OF202401. *(Corresponding author: Yaozhong Pan.)*

Yu Zhu, Yaozhong Pan, and Yao Liu are with the State Key Laboratory of Remote Sensing Science, Institute of Remote Sensing Science and Engineering, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China (e-mail: pyz@bnu.edu.cn).

Tangao Hu is with the Institute of Remote Sensing and Earth Sciences, Hangzhou Normal University, Hangzhou 311121, China, and also with the Zhejiang Provincial Key Laboratory of Urban Wetlands and Regional Change, Hangzhou 311121, China.

Digital Object Identifier 10.1109/TGRS.2025.3628397

I. Introduction

A. Background and Significance of Field Boundary Delineation

AGRICULTURAL parcels (APs) shape agricultural landscapes and are fundamental units of agricultural statistics [1]. They are delineated by field boundaries composed of ridges, pathways, ditches, and so on [2]. The precise delineation of APs is not only the identification of cultivated land areas but also the further subdivision of internal cropland units. Such detailed information is essential for conducting parcel-level assessments, encompassing critical data like crop varieties, growth stages, and phenological characteristics. These assessments support precision agriculture and guide the formulation of government policies. However, traditional AP delineation methods, which depend on labor-intensive manual processes or on-site investigations, are time-consuming. Fortunately, advancements in image resolution and data accessibility have enabled remote sensing-based delineation methods to emerge as efficient and viable alternatives.

B. Review of Existing Delineation Methods

Initial efforts in automated AP delineation were dominated by traditional image processing techniques, which can be broadly categorized into edge- and region-based approaches.

Edge-based methods identify field boundaries by detecting local pixel gradients. Operators like Canny, Sobel, and Laplace were employed [3], [4], [5]. However, these methods operate on low-level features and are susceptible to spurious edges caused by noise or intrafield textural variations. Crucially, they lack semantic context, resulting in an inability to distinguish true field boundaries from other linear features like roads or canals.

In contrast, region-based methods, including techniques like watershed and mean-shift segmentation, aimed to group pixels into homogeneous objects [6], [7], [8]. While effective at generating objects, their effectiveness is limited by a critical dependency on parameter tuning, which often leads to undersegmentation or oversegmentation. Although some hybrid strategies (e.g., CEWS) attempted to combine the strengths of both methods, they generally inherited the limitations of their parent techniques [9], [10], [11], [12].

The constraints of these traditional approaches, particularly their reliance on handcrafted features and sensitivity to

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

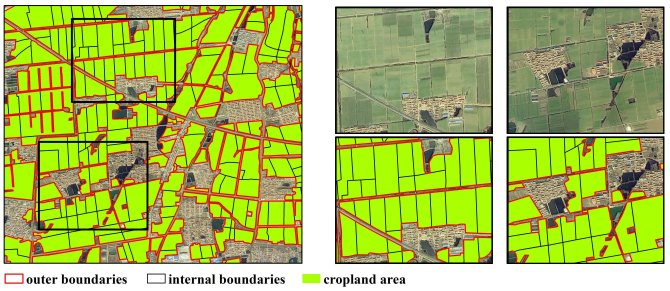

Fig. 1. Visual presentation of field boundaries, which are composed of both external and internal boundaries.

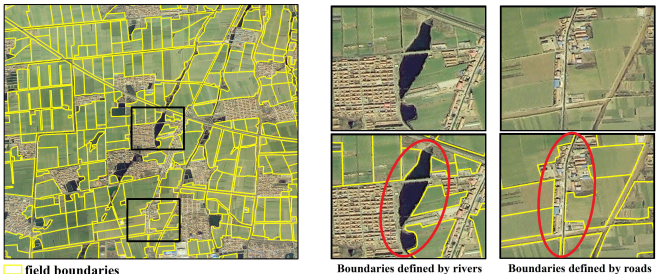

Fig. 2. Illustration of the context-dependent semantics of field boundaries. The red circles highlight how the same linear feature (e.g., a road and river) can transition from field boundaries to nonfield boundaries when they extend beyond the cropland areas.

parameterization, highlighted the need for more robust and automated solutions. This led to a shift toward more powerful deep learning-based methods.

Early deep learning-based delineation studies focused on recognizing cropland region. For instance, Zhang et al. [13] used a modified PSPNet (MPSPNet) model to map high-resolution croplands, achieving overall accuracies exceeding 90% in four provinces in China. Similarly, architectures like DeeplabV3 + and UNet have been used for cropland region recognition [14], [15]. However, these models primarily differentiate cropland from noncropland areas, often at the expense of fine detail, leading to the loss of internal boundaries. This makes them suitable for mapping overall cropland distribution but inadequate for parcel-level analysis.

Recognizing the importance of fine boundary information, subsequent research shifted focus to the direct detection of field boundaries. Field boundaries comprise two types: external boundaries separating cropland from noncropland (red lines, Fig. 1) and internal boundaries defining individual parcels (black lines, Fig. 1). Considering conventional edge operators proved ineffective in complex agricultural scenes [10], [12]. This led to the widespread adoption of deep learning methods for field boundary detection, leveraging their powerful feature description capabilities. These methods can be broadly categorized.

Edge Network Approaches: Generic edge detection networks [e.g., HED, richer convolutional feature (RCF), and DexiNed] have been adapted for field boundary identification [4], [16], [17]. While these models exhibit excellent boundary awareness by fusing multiscale boundary maps, they lack category-specific semantic understanding and often struggle to distinguish true field boundaries from other edges like roads or building outlines.

Semantic Segmentation Network Approaches: This strategy treats field boundaries as a distinct semantic class, thus detecting them semantically. Commonly used architectures include FCN [18], [19], SegNet [1], U-Net [20], [21], [22], and its variants like ResU-Net [14] and R2U-Net [23]. While these networks excel at semantic discrimination, they often face limitations in precise boundary perception. The progressive upsampling in these models can blur fine details, resulting in fragmented boundary delineations.

Multitask Learning (MTL) Approaches: The current state-of-the-art attempts to resolve this conflict by predicting regions and boundaries simultaneously. For example, the ResUNet-a model [24] adds modules and a multitask head to ResU-Net for this purpose. The BsiNet [25] and SEANet [26] models have advanced this by designing independent decoders for region and boundary tasks. The latest approaches, such as TFNet, which integrates FastSAM priors into its encoder for multitask output [27], and HBGNet, which employs a local–global context aggregation strategy [28], continue to push for a better combination of semantics and detail. However, while these MTL frameworks outperform single-task models, they still suffer from two key flaws. First, they often employ parallel, noninteracting branches, failing to use high-level semantic context to explicitly guide and constrain boundary detection. Second, they are typically built on segmentation backbones, which are inherently less suited for precise, crisp boundary localization than dedicated edge networks. This results in boundary maps that are often incomplete or fragmented.

C. Challenges in Field Boundary Detection

The accurate and continuous delineation of field boundaries from remote sensing imagery can be further complicated by several inherent characteristics, which have received limited attention and lack dedicated solutions in prior work.

Complex Composition and Semantics: Field boundaries are not a single feature type but a composite of roads, ditches, uncultivated strips, or subtle texture changes. Furthermore, the semantic meaning of field boundary is context-dependent; for example, a path may be a field boundary inside a farm but irrelevant outside it (as illustrated in Fig. 2). This contextual dependency and compositional diversity make it difficult for models to rely solely on local visual cues for recognition.

Linear Connectivity and Variable Morphology: Field boundaries are linear and connected structures, forming intricate networks. They also exhibit significant morphological variability (curvature and branching patterns). Standard or dilated convolutional kernels, with their fixed receptive fields, struggle to capture these slender and dynamically shaped structures [Fig. 3(a) and (b)]. While deformable convolutions [29] offer adaptability, their discrete sampling points can fail to preserve the linear connectivity essential for boundaries [Fig. 3(c)].

Large Spans and Variable Lengths: Field boundaries often form large, interconnected networks, requiring models to have expansive receptive field to capture this long-range information. Furthermore, multiscale boundary features are also

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

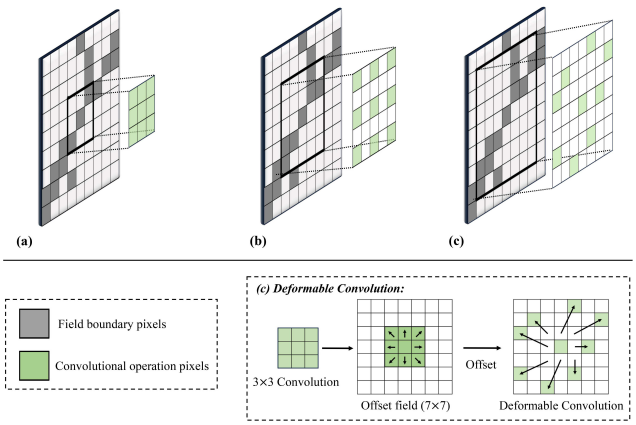

Fig. 3. Convolution operation pixels of different types of kernels, where (a) standard convolution and (b) dilated convolution have fixed convolution operation pixels, while (c) deformable convolution, although capable of adaptively offsetting convolution operation pixels, struggles to adapt to the linearly connected field boundaries due to its discrete operation regions.

crucial, as boundaries can range from short segments to long lines. However, conventional models typically enlarge receptive fields through downsampling, a process that degrades resolution and compromises boundary localization. Therefore, how to effectively capture these large-span, linear, and multiscale boundary features to ensure connectivity remains a challenge.

Boundary Occlusion and Blurring: In high-resolution imagery, features like trees, shadows, or image blur frequently obstruct boundaries. These occluded segments lack distinct boundary characteristics, making it difficult for models to establish a direct relationship between the ambiguous local information and the true boundary. While deep learning models can resolve such image ambiguities or degradations by learning from extensive datasets and contextual information [30], [31], their ability to do so requires extensive sample support, which is often unavailable in real-world applications. Since boundary connectivity is paramount for parcel-level analysis, specific model designs are necessary to guide the model to minimize the impact of occlusions and maximize connectivity. Particularly in data-scarce scenarios, such specialized modules could offer a crucial inductive bias to compensate for the lack of information in limited training samples.

D. Contributions of This Work

To address the aforementioned challenges, this study proposes a high-connectivity and accuracy field boundary network (HCFNet), which integrates a lightweight semantic network with an enhanced edge network based on the following principles.

1) *Task Decomposition With Semantic Gating:* We redefine the problem as detecting "edges within a cropland region." A lightweight semantic network first identifies the cropland extent, and this mask is then used to provide a powerful semantic constraint to a dedicated, high-performance edge network. This synergistic design combines the semantic power of segmentation with the detail awareness of edge detection.
2) *Specializing the Architecture*: We enhance the powerful DexiNed edge network with several modules tailored for field boundaries. These include a linearly connected deformable convolution (LCDConv) to capture fine boundary morphology, nonlocal operations (NLOs) to reinforce the learning of ambiguity boundaries with global context, and stacking multiscale dilated strip convolutions with both fixed and adaptive voids to enrich long-distance features and further address boundary occlusion by explicitly modeling it. Finally, an attention-based module adaptively fuses the multilevel boundary maps.

E. Article's Organization

The remainder of this article is structured as follows. Section II delves into the HCFNet architecture. Section III outlines the datasets, implementation details, and evaluation metrics used. Section IV reports our experimental results, showcasing ablation and comparison studies, and the evaluations of architectural design. Section V offers an in-depth discussion of these results, focusing on the efficacy of our design choices. Finally, Section VI concludes this article and proposes avenues for future research.

II. Proposed Method

A. Overall Network Architecture

The proposed HCFNet is designed to accurately delineate field boundaries by synergistically combining semantic region understanding with specialized edge detection. The overall architecture, depicted in Fig. 4, follows a two-stage sequential process.

Stage 1 (Semantic Cropland Region Task:) This initial stage identifies the broader cropland areas within the input remote sensing imagery. It employs a lightweight encoder–decoder semantic segmentation network based on a VGG architecture. Before generating the final cropland mask, a lightweight spatial group-wise enhance (SGE) module refines the feature maps by reweighting channel importance based on global information. The output of this stage is a binary mask representing cropland versus noncropland areas, which serves as a crucial semantic constraint for the subsequent boundary delineation task.

Stage 2 (High-Connectivity Boundary Task:) This second stage is responsible for the precise delineation of field boundaries. It takes the original image concatenated with the cropland mask from Stage 1 as input. This concatenation explicitly guides the boundary task to focus on edges within the identified cropland areas. The core of this stage is an advanced edge detection network incorporating several novel components designed to enhance boundary connectivity.

B. High-Connectivity Boundary Task

The boundary delineation task is structured around a robust edge detection framework, further augmented with specialized modules to address the challenges of field boundary extraction.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

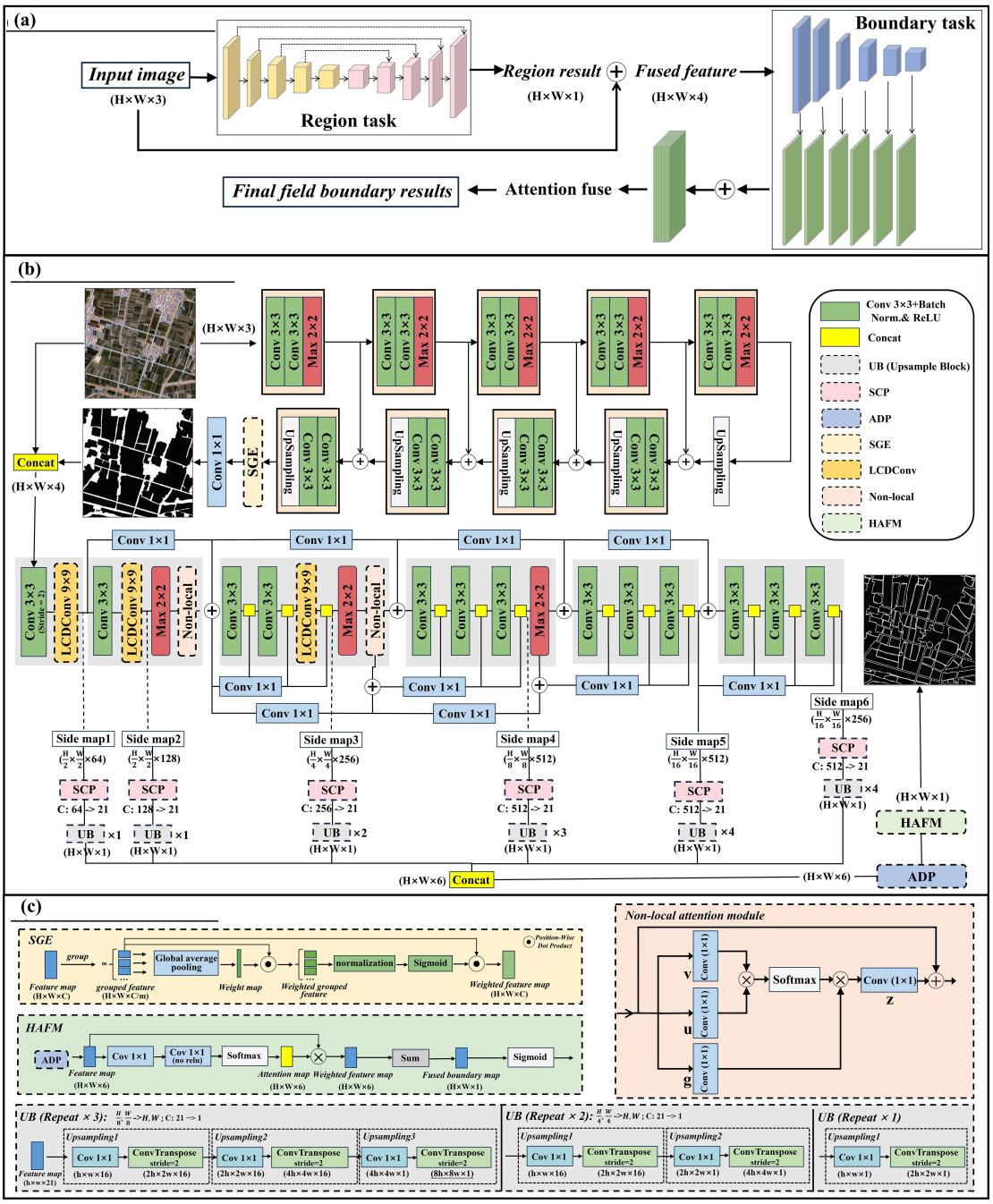

Fig. 4. Illustration of the proposed HCFNet. The figure presents the network from three levels. (a) Overview of the entire framework and data flow; (b) detailed, layer-by-layer architecture of the network; and (c) internal structures of the components.

1) Backbone Edge Network: The foundation of the boundary task is an edge detection network inspired by DexiNed [32], known for its strong inherent boundary perception. This backbone typically employs an encoder–decoder structure with multiple side-output branches from different levels, which are then fused to produce the final boundary map. As shown in Fig. 5, it consists of six Dexi blocks for multiscale feature extraction, upsampling blocks (UBs) to rescale the six intermediate feature maps to their original size, and a final 1×1 convolution to fuse these maps into a single-channel boundary result.

To better adapt this backbone for the high-connectivity field boundary task, we have made the following four improvements.

2) LCDConv: Capturing Boundary Morphology: Existing convolution strategies (standard, dilated, and deformable) are ill-suited for the slender, continuous nature of field boundaries. To address this, we design a more effective LCDConv.

Inspired by deformable convolution, LCDConv introduces offsets to adaptively adjust the sampling region. However, unlike standard deformable convolution, which learns all offsets simultaneously, LCDConv imposes a connectivity

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

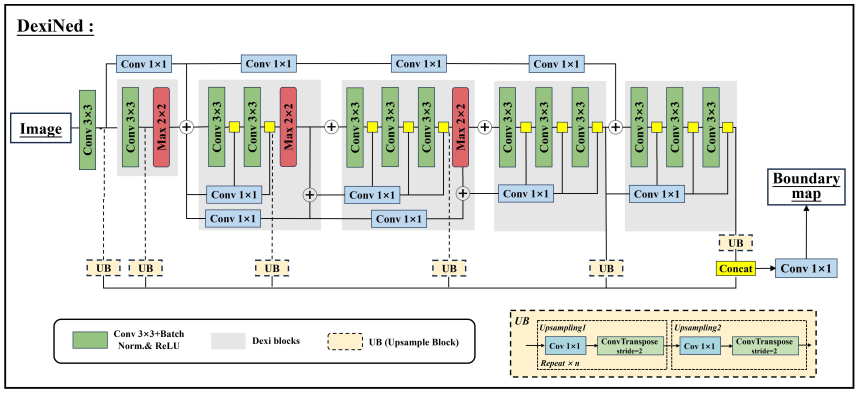

Fig. 5. Architecture of the original DexiNed edge network.

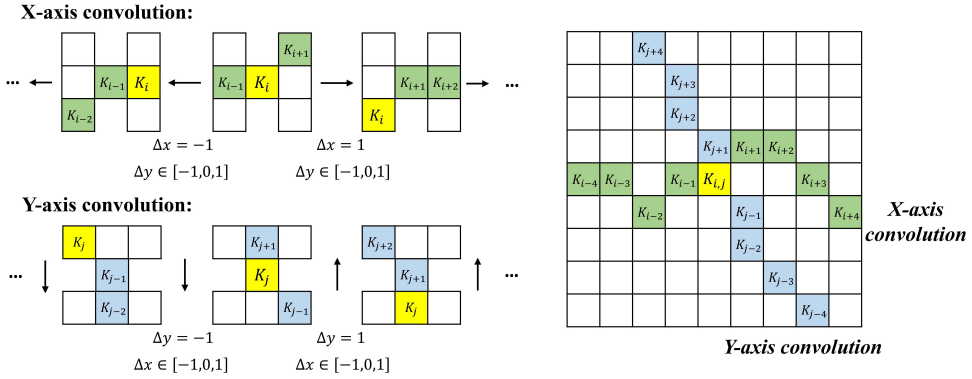

Fig. 6. Example illustrating the process of LCDConv dynamically generating linearly connected convolution regions. The left image shows the step-by-step computation of the next convolution pixel along the X- and Y-axes, while the right image displays a generated 9×9 LCDConv.

constraint. This is achieved through an iterative strategy that calculates the offset for the next sampling point based on the position of the previous point.

Specifically, LCDConv starts from the central position K_i and calculates the sampling locations along the X- and Y-axes sequentially. The next sampling pixel is calculated based on the previous pixel and a trainable offset. For a 9×9 kernel, starting from the central coordinates (X_i, Y_j), the coordinates of the next two convolution pixels along the positive x-axis, K_{i+1} and K_{i+2}, are given by

$$K_{i+1} = \left(i+1, j + \Delta y_{i+1}\right) \tag{1}$$

$$K_{i+2} = \left(i+2, j + \Delta y_{i+1} + \Delta y_{i+2}\right). \tag{2}$$

Thus, all convolution pixels along the x-axis (K_X) and y-axis (K_Y) can be summarized as

$$K_X = \begin{cases} K_{i+c} = \left(i + c, j + \sum_{i}^{i+c} \Delta y\right) \\ K_{i-c} = \left(i - c, j + \sum_{i}^{i-c} \Delta y\right) \end{cases} \tag{3}$$

$$K_Y = \begin{cases} K_{j+c} = \left(i + \sum_{j}^{j+c} \Delta x, j + c\right) \\ K_{j-c} = \left(i - \sum_{j}^{j-c} \Delta x, j - c\right) \end{cases} \tag{4}$$

where $c = \{0, 1, 2, 3, 4\}$ and $\Delta = \{\delta | \delta \in [-1, 1]\}$.

As shown in Fig. 6, this iterative approach allows LCDConv to adaptively form linearly connected sampling regions, thereby covering field boundary pixels more effectively and helping the model capture more complete and accurate linear features. We replace the original standard convolutions in the Dexi blocks with our proposed LCDConv (Fig. 4).

3) NLO: Overcoming Ambiguity With Global Context: Boundaries are often rendered ambiguous by occlusion or blur. Models relying only on local features may fail to establish its connection to the true boundary, especially in data-scarce scenarios. However, these ambiguous patterns often appear repeatedly across the image with a similar visual appearance (Fig. 7). Our insight is that the model can learn to resolve these local ambiguous more effectively by referencing visually similar patterns from other locations within the image.

To guide the model in learning from these long-range dependencies, we introduce a nonlocal operations (NLOs) module [33] into HCFNet. The NLO module calculates the response at each pixel as a weighted average of features from all other pixels in the image. The weights are determined by feature similarity, not proximity. This allows the model to build long-range spatial dependencies, effectively using the global context of the entire image to reinforce its understanding of localized, ambiguous patterns like tree shadows or blur.

Specifically, a nonlocal operation is described as

$$y_i = \frac{1}{C}\left(\sum_{j=1}^{N} f(x_i, x_j) g(x_j)\right) \tag{5}$$

$$g\left(x_j\right) = w_g x_j \tag{6}$$

$$f\left(x_i, x_j\right) = u\left(x_i\right)^T v\left(x_j\right) = \left(w_u x_i\right)^T w_v x_j \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

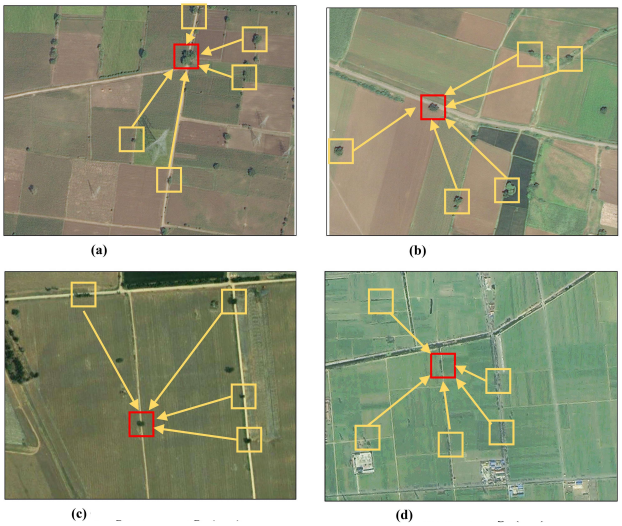

Fig. 7. Some occluded boundaries on different images. Red boxes indicate local windows where boundary occlusion occurs, and yellow boxes highlight similar occluded boundaries in the surrounding areas. (a) UAV image (0.3 m). (b) UAV image (0.5 m). (c) Google Earth image (1 m). (d) Gaofen-1 image (2 m).

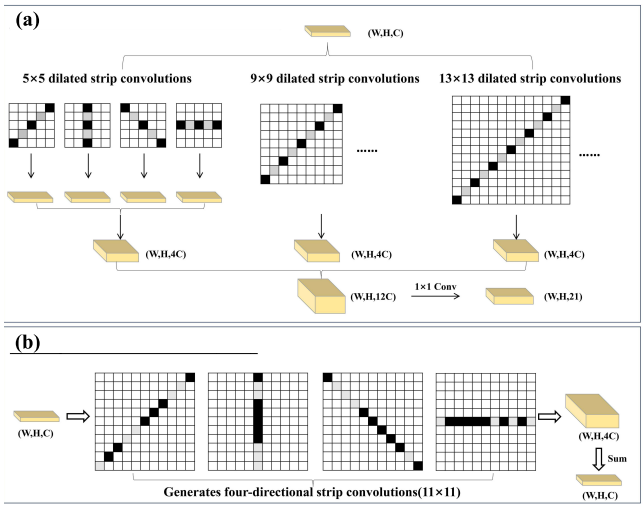

Fig. 8. Architecture of the CRD. The CRD is composed of two core modules. (a) SCP for capturing long-range context and (b) ADP for fine-grained refinement.

where N is the dimension of the input feature map, i is the calculated position index, and j is the index of all possible positions. $x \in R^{N \times C_1}$ is the input feature map and $y \in R^{N \times C_2}$ is the output feature map, where C_1 and C_2 are the number of input and output channels, respectively. x_i,x_j, and y_i are the ith and jth input feature and the ith output feature, respectively. A pairwise function $f(x_i, x_j)$ calculates the feature similarity (weights) between x_i and all possible points x_j, where signals x_i and x_j first pass through their respective 1×1 convolutions for channel adjustment (w_u and w_v), and the dot product is used to evaluate the pairwise relationship. The unary function $g(x_j)$ adopts a 1×1 convolution (w_g) to adjust the input signal xj into target channel number C_2. Finally, y_i calculates the weighted sum of all possible positions x_j and the result is normalized by a factor $C \in R+$. w_u, w_v, and w_g correspond to three 1×1 convolutions ($u, v,$ and g) in Fig. 4.

We then connect the result of the nonlocal operation with an input feature via a residual connection. The residual connection is implemented with an additional 1×1 convolution (w_z) for channel adjustment

$$z_i = w_z y_i + x_i. \tag{8}$$

This residual connection allows the NLO block to be inserted anywhere in the network. For maximal efficiency, we place it after the first two Dexi units to enhance feature representation early in the network.

4) CRD: Receptive Field Expansion and Explicit Occlusion Modeling: To address boundary fragmentation caused by occlusions from trees and shadows, we introduce a novel continuity restoration decoder (CRD). Integrated into the decoding path, the CRD is designed to produce coherent boundary maps by effectively expanding the receptive field and perceiving occluded boundary segments. This is accomplished through two specialized modules: the structured context processor (SCP) for long-range context and the adaptive detail processor (ADP) for fine-grained refinement (Fig. 8).

The guiding principle of the CRD is the explicit modeling of occluded boundaries, which manifest geometrically as discontinuous, "holed" strips. By simulating this gapped structure using a technique called "strip convolution with holes," the CRD learns to infer the underlying boundary continuity. Both modules use this technique but employ different strategies.

a) Structured Context Processor: The SCP operates on the side-output feature maps from the encoder, where semantic information is rich, but spatial resolution is low. Its primary function is to re-establish the large-scale connectivity of boundaries fragmented by regular occlusions. The SCP achieves this through strip convolutions with deterministically generated hole patterns. This strategy of stacking dilated convolutions expands the kernel's receptive field. Furthermore, this strip-based approach is more effective than stacking standard dilated convolutions because it better conforms to the linear morphology of boundaries, making it more beneficial for modeling their long-range connectivity. The SCP is composed of two designs.

Systematic Hole Patterning: We use standard strip convolutions with dilation to create structured gaps in the kernels, thereby enlarging the receptive field and simulating systematic occlusions.

Anisotropic and Multiscale Kernels: The module applies these hole convolutions across four directions (horizontal, vertical, and diagonals) and in parallel branches with varying kernel sizes (e.g., 5×5, 9×9, and 13×13) to robustly model boundaries of diverse orientations and lengths.

By integrating these elements, the SCP's receptive fields can "span" across occluded regions and leverage contextual cues from visible boundary segments to "bridge" gaps on multiscale feature maps, benefiting long-range structural integrity.

b) Adaptive Detail Processor: The ADP is deployed at the final decoding stage to process upsampled feature maps, where finer, more irregular occlusions are visible. The ADP

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

generates four-directional strip convolutions on an 11×11 kernel, but unlike the deterministic patterns in the SCP, the locations of the "holes" in the ADP's kernels are dynamically learned. This allows the kernels to simulate a wide variety of irregular occlusions, enhancing the model's adaptability. The ADP is also implemented as a residual unit for enhanced training stability.

5) HAFM: Adaptive Multiscale Fusion: A key strength of DexiNed backbone is its generation of multiscale side-output boundary maps. However, the optimal way to fuse these maps—balancing fine details from shallow layers with robust context from deep layers—is scene-dependent. Conventional static fusion strategies, like a simple 1×1 convolution, utilize fixed weights and thus lack the adaptability to address this variability.

To solve this, we propose the hierarchical attention fusion module (HAFM). The HAFM replaces static fusion with a dynamic, attention-based weighting mechanism. It employs a subnetwork of 1×1 convolutional layers to analyze the features from all scales and predict attention weights. These weights determine the contribution of each side-output map to the final fusion. This allows the model to adaptively emphasize, for example, fine-grained details when analyzing dense parcels or high-level semantic context when analyzing broad fields. This dynamic approach is critical for achieving robust generalization across diverse agricultural landscapes.

C. Loss Function

The HCFNet is trained by optimizing a combined loss function that addresses both the semantic cropland region task and the boundary delineation task.

Region Task Loss (L_{reg}): For the cropland region segmentation task, a combination of binary cross-entropy (BCE) loss and Dice loss is employed. This hybrid loss handles the potential class imbalance between cropland and noncropland pixels and helps stabilize training [34], [35]. The BCE loss (l_{BCE}) focuses on pixel-wise classification accuracy, while the Dice loss (l_{Dice}) measures the overlap between predicted and ground-truth regions. The combined region loss is

$$l_{\text{BCE}} = \frac{1}{N}\left(-\sum_{n=1}^{N}\left(P_i \log G_i + (1-P_i)\log(1-G_i)\right)\right) \tag{9}$$

$$l_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} P_i \times G_i + \varepsilon}{\sum_{i=1}^{N} P_i^2 + \sum_{i=1}^{N} G_i^2 + \varepsilon} \tag{10}$$

$$L_{\text{reg}} = 0.5 \times l_{\text{BCE}} + 0.5 \times l_{\text{Dice}} \tag{11}$$

where N is the total number of pixels, P_i is the model's prediction for pixel i, G_i is the ground-truth label for pixel i, and ε is a small constant to prevent division by zero.

Boundary Task Loss (L_{bou}): The field boundary delineation task suffers from significant class imbalance (boundary pixels are a small fraction of total pixels). To counteract this, we use a weighted cross-entropy loss for each of the K side-output boundary maps and the final fused map [see (12)]. The weights are inversely proportional to the class frequencies [36]. This is applied as a deep supervision strategy, where the cumulative boundary loss L_{bou} is the weighted sum of the losses from all boundary predictions [see (13)]]

$$L_{\text{bou}}^{k} = \frac{1}{N}\left(-\sum_{n=1}^{N}\left(\frac{|Y_-|}{|Y_+ + Y_-|}\hat{y}_+^k \log\ y_+^k + \frac{|Y_+|}{|Y_+ + Y_-|}\hat{y}_-^k \log\ y_-^k\right)\right) \tag{12}$$

$$L_{\text{bou}} = \sum_{k=1}^{6} w_k L_{\text{bou}}^{k} \tag{13}$$

where k is the level of the boundary predictions; y_+ and y_- are prediction positive pixels and negative pixels, respectively; $\hat{y}_+$ and $\hat{y}_-$ are the label category of them; and $|Y_+|$ and $|Y_-|$ are the number of their pixels, respectively. w_k represents the weight for each corresponding loss term, with the weights set to [0.3, 0.3, 0.4, 0.5, 0.6, 1.0] for k = 1,..., 6, respectively.

Combined Multicomponent Loss: The overall network is trained by optimizing a combined loss function. While the region and boundary delineation are distinct objectives processed sequentially (with the region output informing the boundary input), their respective losses are jointly optimized. An uncertainty-based weighting scheme [37] is used to dynamically balance the contributions of the region loss (L_{reg}) and the aggregated boundary loss (L_{bou}). This technique allows the model to learn the relative importance of each task during training, avoiding manual tuning of loss weights. This approach has been widely applied in various multitask networks [26], [38], [39], [40] and has also been successfully applied to multitask networks for field delineation [41].

The uncertainty-based multitask loss is derived by maximizing the uncertainty likelihood estimation. Let $f^w(x)$ denote the task output given weight w and input x, with model likelihood $P(y|f^w(x))$, and σ^2 is a trainable noise parameter.

The joint likelihood for boundary and region task losses is given by

$$p(y_1, y_2|f^w(x)) = \text{Softmax}\left(f^w(x), \sigma_1^2\right) \cdot \left(f^w(x), \sigma_2^2\right). \tag{14}$$

Taking the logarithm transforms it into a minimization objective $L(w, \sigma_1, \sigma_2)$, which serves as the total loss L_{total} and can be simplified to the form of the following equation:

$$\begin{aligned} L(w, \sigma_1, \sigma_2) &= -\log\ \text{Softmax}\left(f^w(x), \sigma_1^2\right) \cdot \left(f^w(x), \sigma_2^2\right) \\ &= \frac{1}{2\sigma_1^2} L_{\text{reg}}(W) + \frac{1}{2\sigma_2^2} L_{\text{bou}}(W) + \log \sigma_1 \sigma_2. \end{aligned} \tag{15}$$

III. Experimental Setup and Datasets

A. Datasets

To comprehensively evaluate the performance of our proposed HCFNet, we utilized a diverse set of datasets encompassing various geographical regions, sensor types, and agricultural landscape. These include a collected set of high-resolution imagery and a large-scale public benchmark dataset.

1) Collected High-Resolution Image Datasets: To perform targeted validation, ablation studies, and comparative analyses, we assembled a multisource high-resolution dataset. This collection features imagery from diverse agricultural landscapes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

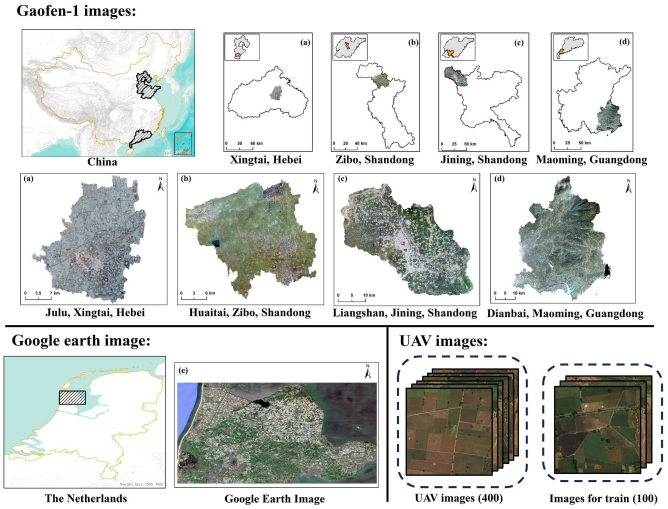

Fig. 9. Overview of the study areas in the high-resolution imagery datasets. (a) Julu County. (b) Huantai County. (c) Liangshan County. (d) Dianbai County. (e) Google Earth images of the study area of the Nertherlands.

and sensor platforms, with all ground truths meticulously prepared. The geographical locations and images are illustrated in Fig. 9.

The collection includes the following equation.

Gaofen-1 (GF-1) Imagery: Four 2-m resolution GF-1 RGB images, all acquired during the growing season over diverse landscapes in China. Three sites, Julu, Huantai, and Liangshan, are located in plains and are dominated by a winter wheat-summer maize rotation. Among them, Julu features small, regular fields, whereas Huantai and Liangshan have large, regular fields. In contrast, the fourth site, Dianbai, is a mountainous area characterized by small, irregular fields dedicated to paddy rice cultivation.

Google Earth Imagery: A scene with 1-m resolution (RGB) covering a large, plain agricultural region in the Netherlands.

UAV Imagery: A set of 400 high-resolution (0.3–0.5 m) images (1024 × 1024 pixels) capturing a wide variety of field types and boundary conditions.

Ground truth for the GF-1 imagery was produced through manual digitization and cross-validated with in situ data. For the Netherlands data, ground truth was sourced from the official Basic Registration of Crop Plots (BRP) database. For the UAV dataset, boundaries were also manually delineated.

For our experiments, the training set was constructed by randomly selecting 25% of the data from each GF-1 and Google Earth imagery scene, combined with 100 images from the UAV dataset. The remaining data from all scenes were used for accuracy validation and testing. For model input, all images in the training set were cropped into patches of 512 × 512 pixels.

2) FTW Dataset: To rigorously test the robustness of HCFNet, we benchmarked it on the fields of the world (FTW) dataset. FTW is a large-scale, standardized dataset designed for global agricultural monitoring research.

Dataset Details: The FTW dataset consists of Sentinel-2 imagery with a 10-m spatial resolution, utilizing the red (Band 4), green (Band 3), blue (Band 2), and near-infrared (NIR, Band 8) bands. It covers an area of 166 293 km^2 across 24 countries. Data are provided as 256 × 256 pixel patches, with each sample location, including imagery from two different time points to capture distinct phenological stages.

Training/Validation/Test Split: Our experiments adhere to the recommended FTW dataset splitting strategy. This involves a blocked random split, and 80% of these blocks are allocated to the training set, 10% to the validation set, and 10% to the test set.

B. Implementation Details

Data Augmentation: Standard data augmentation techniques were applied during training to increase dataset variability. These included random horizontal and vertical flips and random rotations.

Training Parameters: The Adam optimizer [42] was used for gradient descent. The initial learning rate was set to 0.001, and the learning rate was dynamically adjusted using the CosineAnnealingLR strategy. The batch size was 8. Models were trained for up to 150 epochs.

Hardware: All experiments were conducted using a mixed-precision training strategy, accelerated via the hugging face accelerate framework for distributed training on six NVIDIA GeForce RTX 4090 GPUs. Training on our collected high-resolution datasets required approximately 1.5 h, whereas training on the larger FTW dataset took 40.5 h.

C. Evaluation Metrics

The performance of our method was assessed using a multifaceted evaluation protocol. We selected metrics to quantify two aspects of boundary extraction: 1) pixel-level classification accuracy and 2) topological connectivity of the resulting boundaries. For all pixel-based metrics, a tolerance-based matching strategy was employed, where a prediction is counted as correct if it falls within a predefined buffer around a ground-truth pixel. The buffer was set to 1.5 m for high-resolution imagery and 1 pixel (10 m) for the FTW dataset, consistent with IAAO guidelines [43].

1) Pixel-Level Accuracy Metrics: We used a set of standard metrics derived from a confusion matrix (TP, FP, FN, and TN) constructed using the buffer-based matching.

Boundary F1 Score: The harmonic mean of precision and recall was adopted as our primary metric. By focusing on the positive class (boundary pixels), it provides a balanced assessment particularly well-suited for class-imbalanced scenarios such as boundary detection

$$\text{Boundary Precision (BP)} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Boundary Recall (BR)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$\text{Boundary}F1 = 2 \times \frac{\text{BP} \times \text{BR}}{\text{BP} + \text{BR}}. \tag{18}$$

Intersection Over Union (IoU): Also known as Jaccard index, IoU is a fundamental segmentation metric measuring the ratio of the intersection to the union of predicted and ground-truth boundary pixels

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{19}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

TABLE I

ACCURACY METRICS FOR EACH REGION OF THE COLLECTED HIGH-RESOLUTION IMAGE DATASET AND FTW DATASET. THE "1 TIME" AND "2 TIME" SETTINGS REFER TO THE USE OF SINGLE- AND DUAL-PERIOD IMAGERY, RESPECTIVELY

Areas	F1 score	IoU	MCC	GRB (%)
Julu	0.945	0.895	0.929	+10.5%
Huantai	0.961	0.924	0.944	+3.7%
Liangshan	0.969	0.939	0.948	+7.3%
Dianbai	0.886	0.795	0.864	+13.1%
Netherlands	0.975	0.951	0.952	+4.1%
UAV Imagery	0.954	0.912	0.938	+8.2%
Average high-resolution image datasets	0.948	0.903	0.929	+7.8%
Average FTW dataset (1 time)	0.887	0.816	0.873	+11.2%
Average FTW dataset (2 time)	0.899	0.835	0.882	+9.4%

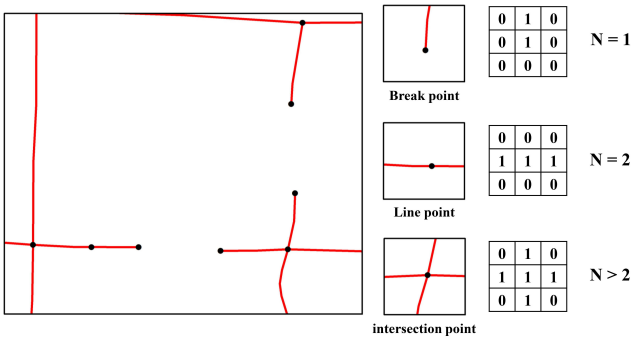

Fig. 10. Principle of breakpoint detection. N denotes the number of boundary (nonzero) pixels in the eight neighborhoods around the point.

Matthews Correlation Coefficient (MCC): A reliable metric that produces a correlation value between −1 (total disagreement) and + 1 (perfect prediction), summarizing the confusion matrix into a single, balanced score

$$\mathrm{MCC} = \frac{(\mathrm{TP} \times \mathrm{TN}) - (\mathrm{FP} \times \mathrm{FN})}{\sqrt{(\mathrm{TP} + \mathrm{FP})\,(\mathrm{TP} + \mathrm{FN})\,(\mathrm{TN} + \mathrm{FP})\,(\mathrm{TN} + \mathrm{FN})}}. \tag{20}$$

2) Topological Connectivity Metric: GRBs: This metric quantifies the fragmentation of the predicted boundary structure. It is calculated by first thinning both prediction and ground truth to single-pixel-wide skeletons and then counting the number of "breakpoints" (pixels with only one neighbor in an eight-connected neighborhood), as illustrated in Fig. 10. A lower growth ratio of breakpoints (GRBs) signifies a more complete and less fragmented boundary network

$$\text{growth ratio} = \frac{\mathrm{BP_{DR}} - \mathrm{BP_{GT}}}{\mathrm{BP_{GT}}} \tag{21}$$

where $\mathrm{BP_{DR}}$ and $\mathrm{BP_{GT}}$ represent the number of breakpoints in the predicted and ground-truth boundaries, respectively. Lower values indicate better connectivity.

IV. RESULTS AND ANALYSIS

A. Results of Field Boundaries Delineation

1) Collected High-Resolution Image Datasets: We first evaluated HCFNet on our collected high-resolution datasets to assess its performance in diverse agricultural landscapes. The quantitative metrics are summarized in Table I, with qualitative visual results provided in Fig. 11.

HCFNet demonstrated stable performance across all test sites. In well-structured plain regions (Julu, Huantai, Liangshan, and Netherlands), the model consistently achieved $F1$ score above 0.94. The highest accuracy was recorded in the Netherlands site (1-m Google Earth imagery), with an $F1$ score of 0.975. The model's robustness was further validated in the challenging mountainous terrain of Dianbai, where it maintained an $F1$ score of 0.886 by successfully discriminating agricultural areas from adjacent, spectrally similar forest regions. On the ultrahigh-resolution UAV data, the model also yielded a strong average $F1$ score of 0.954. Visually, the results show that our method effectively handles occlusions caused by dense vegetation, a common artifact in such imagery.

Furthermore, across all sites, HCFNet produced continuous boundaries, reflected by a very low average GRB of just + 7.8%. This provides quantitative evidence that our architectural design for connectivity effectively preserves the topological integrity of the field networks.

2) FTW Dataset: To assess the robustness of HCFNet on a global scale, we benchmarked it on the FTW dataset. The results are reported in Table I.

Our analysis indicates that leveraging the provided bi-temporal Sentinel-2 imagery was crucial, significantly enhancing the completeness and accuracy of the delineated boundaries over single-date inputs. The primary performance bottleneck on FTW is its 10-m resolution, which is often insufficient to capture the fine, intraparcel boundaries. This challenge is exacerbated by the nature of the ground truth. For many regions, particularly in Europe, the FTW dataset uses official parcel data that was originally delineated from high-resolution sources like Google Earth. This data-label mismatch between the coarse-resolution imagery and the fine-resolution labels leads to a reduction in accuracy. Despite this limitation, visual assessment reveals that our model generates clear parcel maps that accurately represent the field division patterns visible at a 10-m resolution.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

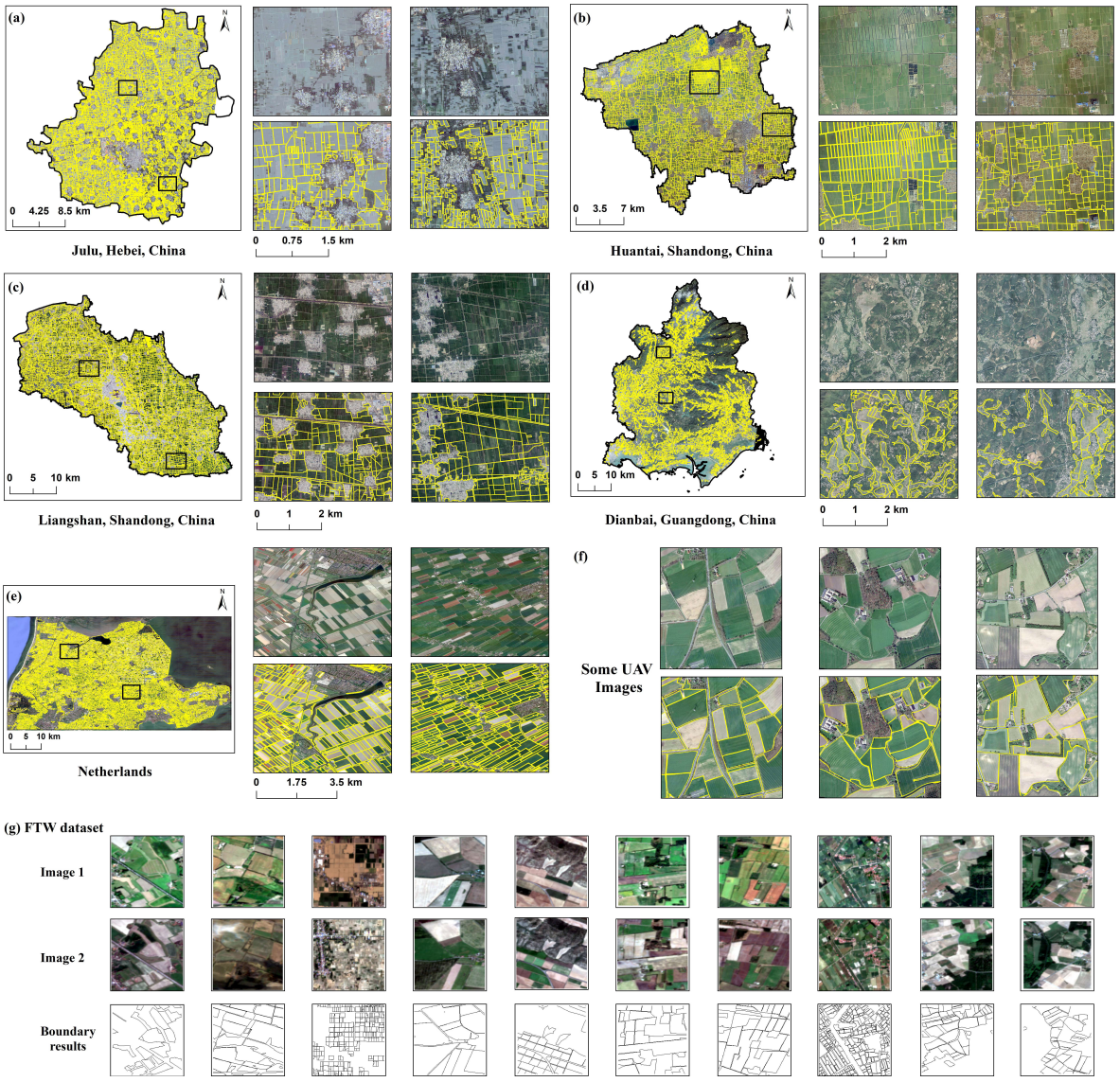

Fig. 11. Delineated field results in the study areas. (a) Julu. (b) Huaitai. (c) Liangshan. (d) Dianbai. (e) Netherlands. (f) Some UAV images. (g) Some images of the FTW dataset.

B. Ablation Experiment Analysis

To quantitatively and qualitatively validate the individual contribution of each proposed component within HCFNet, we conducted a series of ablation experiments. Starting from the full model, we systematically ablated key modules and measured the performance impact. All experiments were run under identical training conditions until convergence. Quantitative results are detailed in Tables II and III, with visual examples in Fig. 12.

Semantic Guidance (Region Task): Removing the region segmentation task (the "Non-RST" model) is equivalent to performing boundary detection using only the edge branch. This led to a significant performance drop across all metrics on the high-resolution dataset (e.g., $F1$ decreased from 0.948 to 0.879) and a dramatic increase in boundary fragmentation (GRB from + 7.8% to + 32.4%). A similar performance gain from the region task was observed on the FTW dataset. This result confirms the crucial role of high-level semantic constraints in filtering irrelevant edges (e.g., from roads or rivers) and improving the semantic accuracy of the final boundaries.

Boundary Morphology (LCDConv): Replacing LCDConv with standard convolutions ("Non-LCDConv") resulted in a notable performance decrease on the high-resolution dataset ($F1$ to 0.913 and GRB to + 10.2%), with a similar accuracy gain observed on the FTW dataset. This underscores the benefits of LCDConv's design. Its elongated, linear receptive field improves boundary connectivity by proactively bridging minor gaps. Furthermore, LCDConv produces thinner, more precise boundary lines that better conform to ground truth, which is advantageous for downstream vectorization. Activation maps in Fig. 13(b) further corroborate this, showing that the model with LCDConv learns to focus its attention more precisely on the slender, linear geometry of field boundaries compared to the model with standard convolutions.

Long-Range Context (NLO): The NLO module is designed to leverage long-range contextual information, guiding the model to reinforce its learning on ambiguous cases like

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

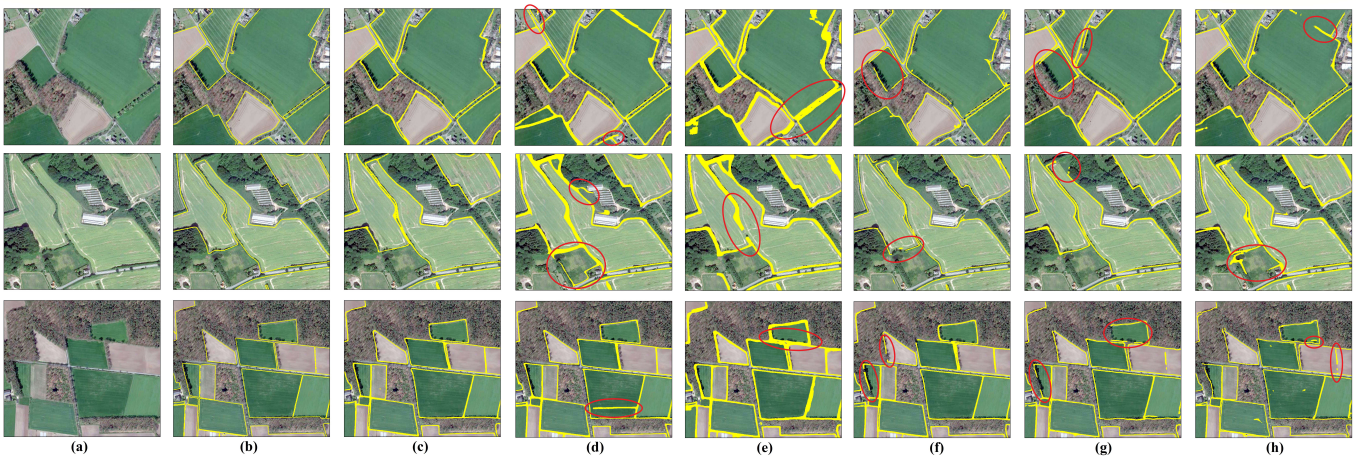

Fig. 12. Visual comparison of delineation results from our full HCFNet model and its ablated variants on a high-resolution dataset. The impact of removing each key component is shown. (a) Images. (b) Ground truths. (c) Our HCFNet. (d) Non-RST. (e) Non-LCDConv. (f) Non-NLO. (g) Non-CRD. (h) Non-HAFM.

TABLE II

AVERAGE ACCURACY METRICS OF ABLATION MODELS IN THE COLLECTED HIGH-RESOLUTION IMAGE DATASETS. ITALICS INDICATE OUR MODEL RESULTS, BOLD INDICATES THE HIGHEST ACCURACY, AND UNDERLINED INDICATES THE SECOND-BEST RESULTS

Ablation Model	F1 score	IoU	MCC	GRB
HCFNet-Full	***0.948***	***0.903***	***0.929***	***+7.8%***
Non- RST	0.879	0.801	0.857	+32.4%
Non-LCDConv	0.913	0.851	0.893	+10.2%
Non-NLO	0.935	0.885	0.907	+15.8%
Non-CRD	0.929	0.874	0.901	+16.3%
Non-HAFM	0.897	0.829	0.874	+18.5%

TABLE III

AVERAGE ACCURACY METRICS OF ABLATION MODELS IN THE FTW DATASETS. ITALICS INDICATE OUR MODEL RESULTS, BOLD INDICATES THE HIGHEST ACCURACY, AND UNDERLINED INDICATES THE SECOND-BEST RESULTS

Ablation Model	F1 score	IoU	MCC	GRB
HCFNet-Full	***0.899***	***0.835***	***0.882***	***+9.4%***
Non- RST	0.832	0.748	0.810	+26.1%
Non-LCDConv	0.881	0.809	0.864	+12.1%
Non-NLO	0.890	0.822	0.874	+10.4%
Non-CRD	0.895	0.827	0.877	+10.4%
Non-HAFM	0.813	0.716	0.798	+20.7%

occlusions. This guidance proved effective on our small-scale, high-resolution dataset, improving accuracy and connectivity (GRB from + 15.8% to + 7.8%). As illustrated by the activation map for a single occluded pixel [Fig. 13(a)], the NLO module directs the model's attention to structurally similar occluded patterns at distant locations, thereby reinforcing a robust occlusion representation. However, on the data-rich FTW benchmark, which provides extensive examples of such cases, a standard deep network can learn these representations without NLO, rendering the explicit guidance of NLO redundant. Consequently, its addition offered a negligible gain on FTW. Therefore, the inclusion of NLO represents a tradeoff that should be weighed against the training data's scale.

Occlusion Modeling (CRD): The CRD further enhances performance by explicitly modeling occluded boundaries at multiple scales. On our high-resolution dataset, removing the CRD ("Non-CRD") led to a drop in all accuracy metrics and an increase in breakpoints, confirming the efficacy of this hierarchical, occlusion-aware strategy. Activation maps [Fig. 13(c)] show that the CRD is particularly effective in areas with dense vegetation, helping the model perceive a more continuous and well-defined boundary. Similar to NLO, the CRD provides the largest benefit on smaller dataset (collected high-resolution dataset). On large datasets like FTW, its contribution diminishes as the network learns to handle occlusions through exposure to vast data.

Feature Fusion (HAFM): We replaced the HAFM with a simple 1×1 convolution for feature fusion ("Non-HAFM"). This resulted in a substantial drop in accuracy ($F1$ to 0.897 and GRB to + 18.5% in the collected dataset). Unlike a fixed-weight 1×1 convolution, the HAFM adaptively adjusts the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

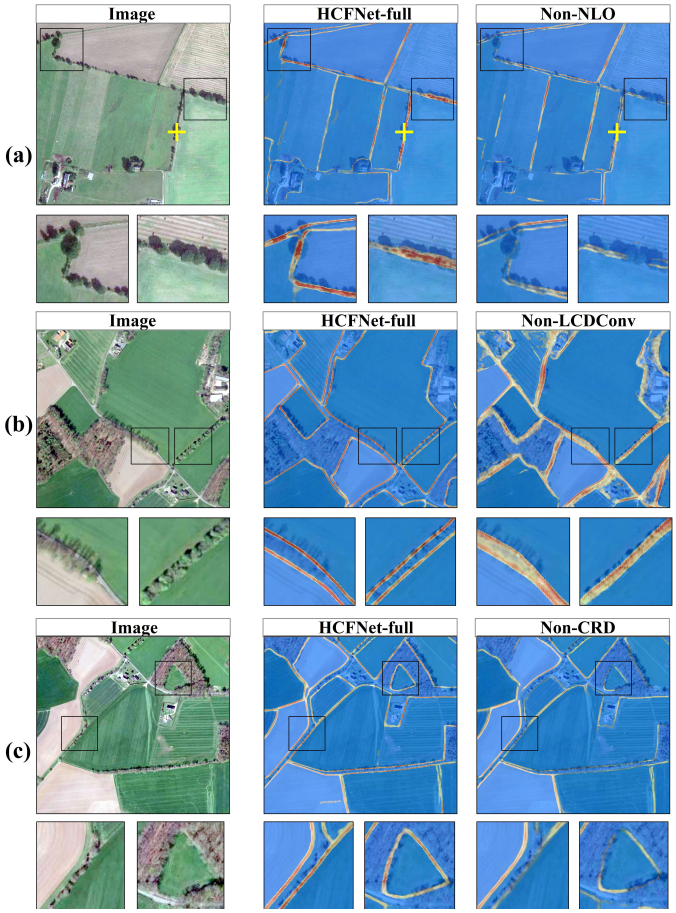

Fig. 13. Ablation study analysis using Grad-CAM heatmaps in UAV images. The figure compares the model's attention with modules removed. (a) Without the NLO module, (b) without the LCDConv module, and (c) without the CRD module. The analysis in (a) is a point-specific heatmap for a vegetation-occluded pixel, while (b) and (c) are class-level heatmaps for the "Field Boundary" class.

weight distribution during the fusion of multiscale features. This adaptability is crucial for generalization across diverse landscapes, making it a universally beneficial component, as demonstrated by its strong performance gains on both datasets.

C. Comparison Experiment Analysis

To benchmark the performance of HCFNet, we conducted a comparative analysis against several commonly employed models for field boundary detection. All models were trained and tested under identical conditions, ensuring full convergence of the training loss (Fig. 14). The quantitative results on both datasets are presented in Tables IV and V, with visual examples in Fig. 15. The competitor models are described as follows.

1) *RCF Edge Model [36]:* An edge detection model that improves upon HED by using RCFs to fuse multiscale boundary information. It has been previously applied to field boundary extraction [17].
2) *SegNet:* A classic deep encoder–decoder segmentation network previously used for detecting field boundaries from high-resolution WorldView-3 images [1].
3) *ResU-Net:* An enhanced U-Net variant that replaces standard convolutional blocks with residual blocks, improving feature representation for boundary detection [14].
4) R2U-Net: An advanced U-Net architecture incorporating recurrent residual blocks, which has shown superior performance over ResU-Net for boundary detection from Sentinel-2 images [23].
5) MPSPNet: An improved PSPNet architecture that enhances shallow feature capture for better detail preservation in cropland extraction [13]. A recent study further adapted this model to multitask prediction and provided effective detection for field boundaries [44].
6) *ResUNet-a [45]:* A multitask U-shaped network that integrates ASPP and PSP pooling modules. It simultaneously outputs region and boundary predictions and is considered a state-of-the-art method [46].
7) *BsiNet:* A recent multitask model derived from PsiNet that incorporates a spatial attention module and uses distinct decoding paths to output cropland areas and boundaries. This model prioritizes region predictions as field results [25].
8) *SEANet:* A recently proposed multitask network combining RCF principles with a parallel dual-branch structure to simultaneously predict regions and boundaries, demonstrating strong edge-aware capabilities. It also prioritizes region predictions as results [26].
9) *HBGNet:* A multitask network featuring a hierarchical, dual-branch architecture where a dedicated boundary branch uses Laplace convolution to extract low-level edge semantics to refine the main parcel extraction branch [28].

As indicated in Tables IV and V, our model outperforms all comparative models in boundary accuracy and connectivity. Specifically, Fig. 15 reveals that single-task edge detectors like RCF struggle with semantic discrimination. While effective at capturing all edge-like features, RCF cannot distinguish field boundaries from other edges, leading to numerous false positives in residential areas (e.g., building edges and internal roads). The resulting poor precision severely suppressed its $F1$ score (0.828 on the high-resolution image dataset). This limitation was also evident on the FTW dataset, where RCF achieved an F1 score of 0.819.

Conversely, single-task segmentation models like SegNet suffer from inadequate boundary awareness, producing fragmented outputs and often failing to detect fine internal field boundaries. The U-Net-based architectures (ResU-Net and R2U-Net) offer stronger feature representation, yet they still exhibit significant boundary fragmentation (average $F1$ scores of 0.854 and 0.861). MPSPNet, by retaining more shallow features, improves detail rendition but still falls short in semantic differentiation and connectivity, with a high breakpoint increase of + 52.2%.

The multitask models show improved performance. ResUNet-a delivers high-quality results than MPSPNet but still struggles with fine-detail perception, overlooking some internal boundaries, and misidentifying noncropland edges. BsiNet and SEANet significantly surpass ResUNet-a by designing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

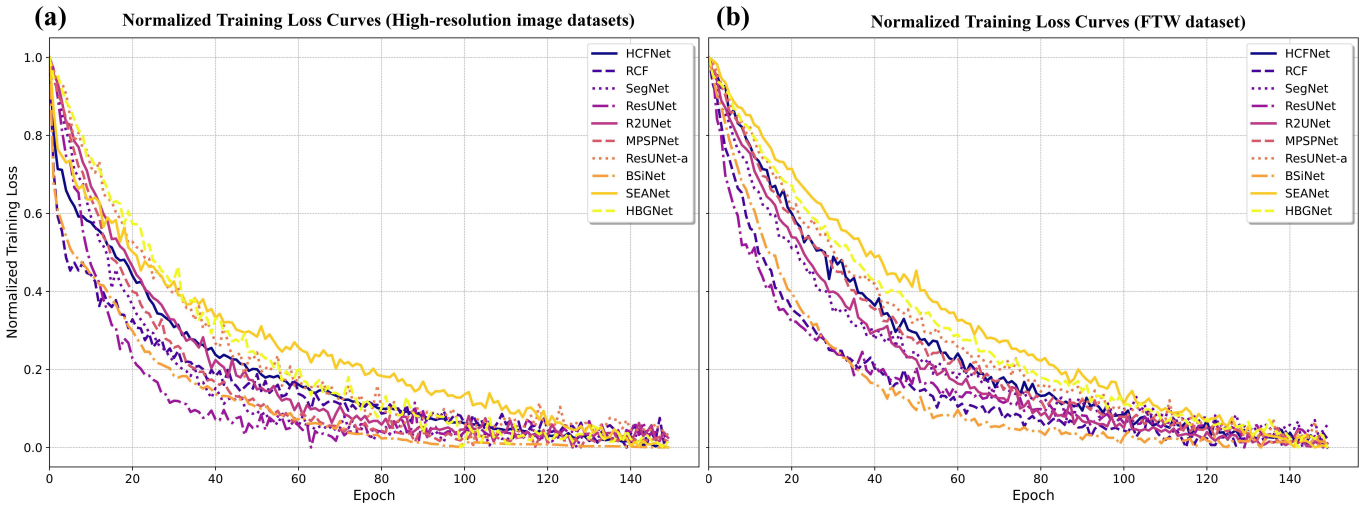

Fig. 14. Training loss curves for different models on (a) high-resolution image dataset and (b) FTW dataset. Since each model was trained using the loss function recommended in its original publication, the absolute scales of the loss values vary significantly. Therefore, all curves have been normalized to the [0,1] range to compare convergence trends fairly.

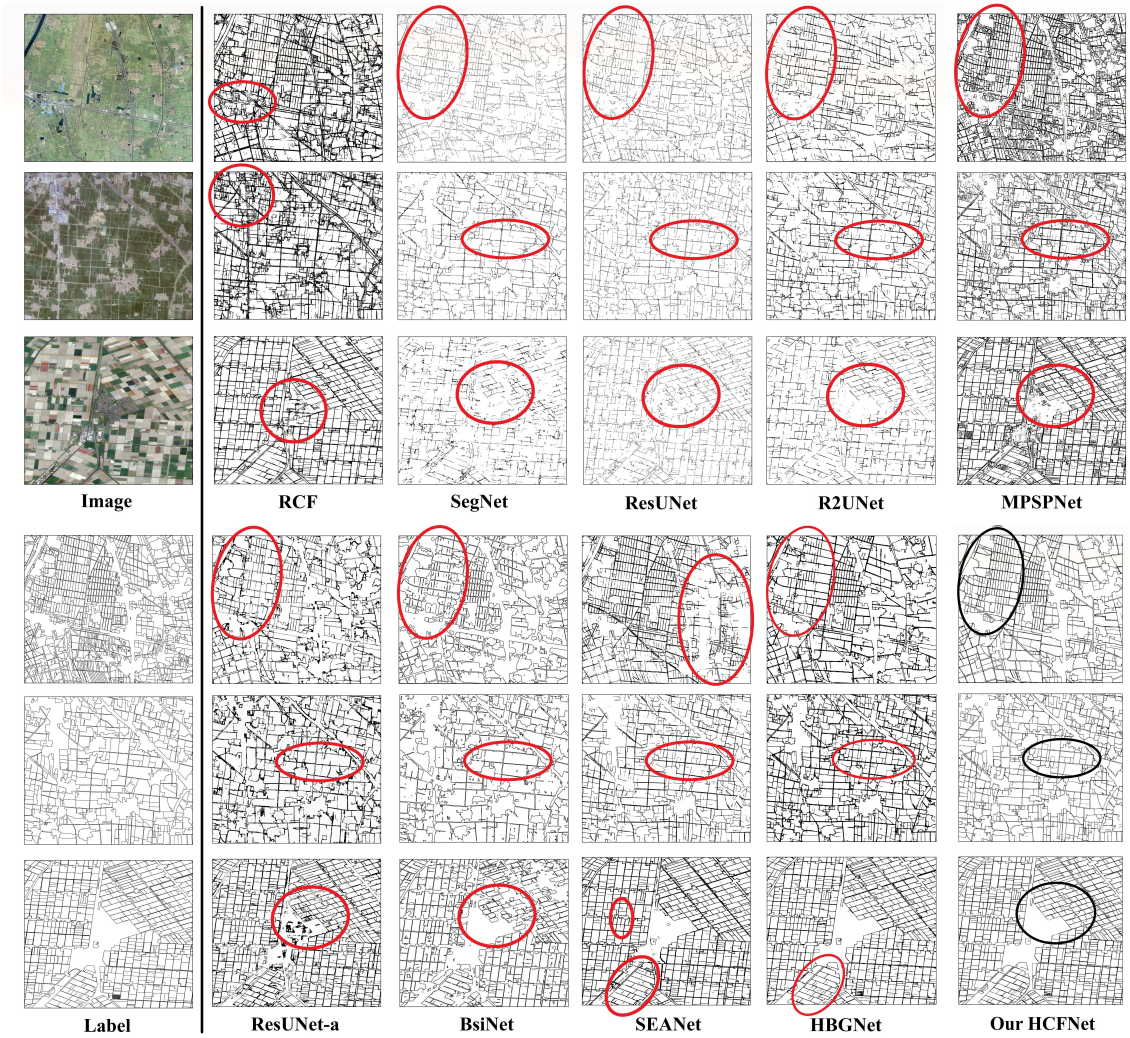

Fig. 15. Detection results of comparison networks. The red circles indicate areas where the comparison networks made incorrect identifications, while black circles show the corresponding results in our network.

independent decoder branches, but their parallel design for region and boundary tasks lacks a mechanism for mutual reinforcement, limiting the role of region-based semantic constraints. Even SEANet, with its bolstered boundary awareness,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

AVERAGE ACCURACY METRICS OF COMPARISON MODELS IN THE COLLECTED HIGH-RESOLUTION IMAGE DATASETS. ITALICS INDICATE OUR MODEL RESULTS, BOLD INDICATES THE HIGHEST ACCURACY, AND UNDERLINED INDICATES THE SECOND-BEST RESULTS

Comparison Model	F1 score	IoU	MCC	GRB
Our HCFNet	***0.948***	***0.903***	***0.929***	***+7.8%***
RCF	0.828	0.737	0.808	+81.7%
SegNet	0.835	0.746	0.816	+85.1%
ResUNet	0.854	0.773	0.834	+71.6%
R2UNet	0.861	0.784	0.844	+67.9%
MPSPNet	0.891	0.817	0.867	+52.2%
ResUNet-a	0.918	0.858	0.881	+41.5%
BsiNet	0.922	0.863	0.885	+33.3%
SEANet	0.930	0.877	0.894	+22.4%
HBGNet	<u>0.936</u>	<u>0.885</u>	<u>0.902</u>	<u>+14.8%</u>

TABLE V

AVERAGE ACCURACY METRICS OF COMPARISON MODELS IN THE FTW DATASETS. ITALICS INDICATE OUR MODEL RESULTS, BOLD INDICATES THE HIGHEST ACCURACY, AND UNDERLINED INDICATES THE SECOND-BEST RESULTS

Comparison Model	F1 score	IoU	MCC	GRB
Our HCFNet	***0.899***	***0.835***	***0.882***	***+9.4%***
RCF	0.819	0.723	0.780	+35.1%
SegNet	0.844	0.754	0.817	+29.2%
ResUNet	0.856	0.769	0.831	+25.1%
R2UNet	0.861	0.775	0.838	+22.4%
MPSPNet	0.871	0.787	0.849	+18.3%
ResUNet-a	0.882	0.804	0.862	+15.7%.
BsiNet	0.879	0.798	0.857	+13.2%
SEANet	0.887	0.812	0.866	+12.1%
HBGNet	<u>0.889</u>	<u>0.816</u>	<u>0.873</u>	<u>+10.9%</u>

TABLE VI

FLOATING-POINT OPERATIONS PER SECOND (FLOPS) AND THE PARAMETERS OF SOME RECENT MODELS (G:10^9 AND M: 10^6)

Methods	SegNet	R2UNet	MPSPNet	ResUNet-a	BsiNet	SEANet	*HBGNet*	*HCFNet*
FLOPs	286.1 G	152.9 G	163 G	73.9 G	13.4 G	208.2 G	*58.9 G*	*55.4 G*
Parameters	29.5 M	39.1 M	46.7 M	134.7 M	7.9 M	28.8 M	*30.5 M*	*44.5 M*

still shows notable fragmentation ($F1$ score of 0.93 and GRB of + 22.4% in the high-resolution dataset). While HBGNet improves performance with branch interaction, its information flow is unidirectional. Low-level boundary features refine the region task, but it lacks the reciprocal feedback from the semantic-related region task to the boundary branch. This limits the semantic accuracy of the extracted boundaries, leading to weaker boundary performance than our model on both datasets.

In contrast, our HCFNet model is founded on the principle of defining field boundaries as "edges within cropland areas." It leverages the region task output to directly impose high-level semantic constraints on the boundary task. Built upon the robust DexiNed framework and enhanced with four specialized modules, HCFNet demonstrates superior boundary perception. It delivers accurate and continuous results, fully capturing fine internal boundaries, achieving an average F1 score of 0.948 and significantly reducing breakpoints across all experiments.

Furthermore, as presented in Table VI, HCFNet is computationally efficient. It has lower floating-point operations per second (FLOPs) than most recent models (SegNet, R2U-Net, MPSPNet, ResUNet-a, SEANet, and HBGNet) and a comparable number of parameters.

D. Analysis of Backbone Network Combinations

Our HCFNet model integrates a lightweight VGG-based semantic network and an enhanced DexiNed edge network. To validate the optimality of this combination, we investigated the effects of varying the baseline network architectures and their respective depths (Tables VII and VIII).

1) Region Branch: Impact of Network Depth: We tested VGG-based region backbones of varying depths. The results (Table VII) revealed a clear point of diminishing returns. While deeper networks slightly improved region segmentation scores, these improvements had a negligible effect on the final boundary accuracy beyond a depth of five layers. Considering

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

NETWORK COMPLEXITY (FLOPS), PARAMETER COUNT, AVERAGE REGION ACCURACY, AND FINAL BOUNDARY ACCURACY IN DIFFERENT COMPARISON REGION NETWORKS (G:10^9 AND M: 10^6)

Different region network	FLOPs	Parameters	Average region accuracy (F1)	Average final boundary accuracy (F1)
Our region model (5 layers) + boundary	*13.2+42.2 G*	*7.9+36.6 M*	*0.862*	*0.948*
Different network depths:				
Our region model (4 layers) + boundary	9.1+42.2 G	4.2+36.6 M	0.836	0.932
Our region model (6 layers) + boundary	17.4+42.2 G	31.5+36.6 M	0.870	0.949
Our region model (7 layers) + boundary	23.7+42.2 G	126+36.6 M	0.879	0.950
Different network architectures (ResNet50-backbone):				
ResU-Net + boundary	63.6+42.2 G	28.7+36.6 M	0.889	0.949
R2U-Net + boundary	153+42.2 G	39.1+36.6 M	0.897	0.951
DeeplabV3+ + boundary	135+42.2 G	41.5+36.6 M	0.903	0.952
MPSPNet + boundary	163+42.2 G	46.7+36.6 M	0.907	0.953

TABLE VIII

NETWORK COMPLEXITY (FLOPS), PARAMETER COUNT, AVERAGE REGION ACCURACY, AND FINAL BOUNDARY ACCURACY IN DIFFERENT COMPARISON REGION NETWORKS (G:10^9 AND M: 10^6)

Different boundary network	FLOPs	Parameters	Average final boundary accuracy (F1)	Average growth ratio of breakpoint number
Region + our boundary model (6 Dexi)	*13.2+42.2 G*	*7.9+36.6 M*	*0.948*	*+7.8%*
Different network depths:				
Region + our boundary model (5 Dexi)	13.2+37.8 G	7.9+31.3 M	0.935	+13.5%
Region + our boundary model (7 Dexi)	13.2+46.8 G	7.9+42.1 M	0.950	+7.6%
Region + our boundary model (8 Dexi)	13.2+52.0 G	7.9+49.0 M	0.951	+7.6%
Different network architectures:				
Region + HED	13.2+32.6 G	7.9+14.7 M	0.915	+28.0%
Region + RCF	13.2+35.4 G	7.9+15.6 M	0.921	+22.4%
Region + BDCN	13.2+36.6 G	7.9+16.3 M	0.930	+17.2%
Region + Pidinet	13.2+45.6 G	7.9+42.3 M	0.939	+13.4%

the higher FLOPs of deeper models, a five-layer network offers the optimal tradeoff between efficiency and performance.

Impact of Network Architecture: We then replaced our lightweight VGG with more powerful segmentation networks, including ResUNet, R2U-Net, DeepLabV3 +, and MPSP-Net, which are applied in cropland region detection [13], [15], [46], [47], with greater complexity. While these models achieved higher region accuracy, they again failed to produce meaningful improvement in the final boundary results.

In conclusion, these experiments consistently show that the semantic guidance from the region task reaches an accuracy plateau. Once the region mask is "good enough" (a region $F1$ score of ~0.86 in our case), making it better does not help the boundary network. Therefore, our choice of a lightweight, five-layer VGG network provides necessary semantic context with maximum computational efficiency.

2) Boundary Branch: We similarly analyzed the boundary network to confirm the optimal depth and architecture.

Impact of Network Depth: We found that the performance of our enhanced DexiNed backbone plateaued at six Dexi modules (Table VIII), making this the most efficient configuration.

Impact of Network Architecture: To prove the superiority of the DexiNed foundation, we conducted a rigorous comparison. We took four other leading edge-detection backbones (HED, RCF, BDCN, and Pidinet) and equipped each one with our full suite of proposed enhancements (LCDConv, NLO, CRD, and HAFM). As shown in Table VIII, even when augmented with our custom modules, none of these alternative backbones could match the performance of our DexiNed-based HCFNet, either in terms of accuracy ($F1$) or connectivity (GRB).

In conclusion, this comparison confirms that the DexiNed architecture possesses an intrinsic superiority in boundary perception that is critical for this task. Its powerful foundation, when combined with our specialized modules, creates the optimal model for high-connectivity boundary delineation.

E. Result-Level Versus Feature-Level Fusion

The HCFNet model incorporates an additional region task to impose semantic constraints, channeling the region task's results directly into the subsequent boundary task. Beyond this result-level fusion, feature-level fusion is another common method for multitask fusion (Fig. 16). To determine which method offers more effective semantic constraints, we conducted comparative experiments between these two fusion strategies.

As presented in Table IX, result-level fusion not only exhibits lower complexity than feature-level fusion but also

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
NETWORK COMPLEXITY (FLOPS), PARAMETER COUNT, AVERAGE BOUNDARY ACCURACY (F1 SCORE), AND AVERAGE GRB NUMBER IN NETWORKS WITH DIFFERENT FEATURE FUSION METHODS (G:10^9 AND M: 10^6)

Different fusion methods	FLOPs	Parameters	Average final boundary accuracy (F1)	Average growth ratio of breakpoint number
Result-level fusion	***55.4 G***	***44.5 M***	***0.948***	***+7.8%***
Feature-level fusion	63.1 G	55.3 M	0.942	+10.9%

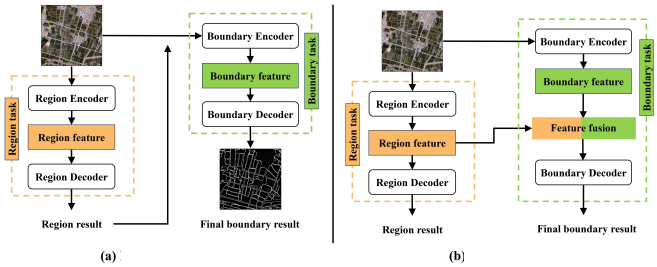

Fig. 16. Two fusion strategies for the feature of region and boundary tasks. (a) Result-level fusion strategy that directly utilizes region task results. (b) Feature-level fusion strategy that fuses region task features with boundary task features.

achieves higher final boundary accuracies and fewer breakpoints. Although feature-level fusion enables the boundary task to leverage richer regional signals, the simultaneous decoding of boundary and regional features increases the decoding complexity and overall model burden. Moreover, the boundary task primarily relies on region results for semantic constraints; multiscale regional features provide little additional value and may even impair the boundary decoder's ability to process boundary features, degrading performance. In contrast, result-level fusion directly utilizes high-quality region masks to deliver semantic constraints to the boundary task. By processing signals independently and linking tasks sequentially, each task can maximize its own performance and achieve complementary functionality, resulting in more accurate identification of field boundaries.

F. Architectural Placement of Specialized Modules

In the HCFNet model, the four performance-enhancement modules serve distinct roles. CRD and HAFM are fixed within the boundary decoding section, tasked with optimizing and fusing multiscale boundary maps. Conversely, LCDConv and NLO reside in the encoding section to refine feature extraction, with their positions adjustable. To determine their optimal placement for maximizing performance, we conducted validation experiments.

1) Positioning LCDConv for Fine-Detail Perception: The LCDConv module is engineered to capture fine, linear boundary structures. Our experiments tested its placement across the six Dexi encoder blocks. The results in Table X confirm our hypothesis: LCDConv is most effective in the early-to-midstages of the encoder (blocks 1–3).

Extending LCDConv to deeper modules (4th, 5th, or 6th), where feature maps are downsampled, yielded no significant accuracy gains. Moreover, shifting the application of LCDConv to deeper positions—such as only in the 2nd, 3rd, and 4th modules—reduced accuracy. Its strength lies in operating on high-resolution features where the slender, connective morphology of boundaries is most apparent. Therefore, we limit its application to the first three Dexi blocks to maximize its impact without adding unnecessary complexity to deeper layers.

2) Positioning NLO for Optimal Contextual Guidance: The NLO module's function is to build long-range dependencies to help the network understand ambiguous regions like occlusions. Its optimal placement requires a careful balance. It needs features that are rich enough to establish meaningful long-range correlations, but it also needs sufficient downstream layers to act upon its guidance.

Our results (Table X) show that placing NLO too early (after Dexi block 1) is ineffective, as the features are too primitive. Conversely, placing it too late diminishes its impact, as there are fewer subsequent layers to influence. We found the optimal position to be after the 2nd and 3rd Dexi blocks. This mid-network placement allows the NLO module to refine well-developed feature maps and effectively guide the deeper half of the network in learning robust representations for challenging, occluded boundaries.

V. DISCUSSION

A. Impact of Region Semantics on Boundary Delineation

A challenge in automated delineation is the inherent tradeoff between preserving fine detail and maintaining correct semantic context. Our work resolves this dilemma by decomposing the delineation process into distinct region and boundary tasks. Our sequential framework uses the output of region task as a "semantic gate" for the boundary task. On a practical level, this means that only edges located within preidentified cropland areas are considered, effectively filtering out noise from roads, buildings, and other nontarget features. More fundamentally, this approach reshapes the feature space itself, forcing the model to learn more discriminative representations for boundaries.

To empirically validate this hypothesis, we used t-SNE [48] to visualize the feature space of different models, plotting 4000 field boundary and 4000 nonfield boundary feature points (Fig. 17). The analysis clearly shows that single-task models struggle to distinguish between the two classes, resulting in heavily overlapping feature clusters. In contrast, our multitask approach achieves superior feature separability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

TABLE X

NETWORK COMPLEXITY (FLOPS), PARAMETER COUNT, AVERAGE BOUNDARY ACCURACY (F1 SCORE), AND AVERAGE GRB NUMBER IN NETWORKS WITH DIFFERENT FEATURE FUSION METHODS (G:10^9 AND M: 10^6)

Module Location	FLOPs	Parameters	Average final boundary accuracy (F1)	Average growth ratio of breakpoint number
Location of LCDConvs:				
In Dexi 1,2,3 (our setting)	***55.4 G***	***44.5 M***	*0.948*	*+7.8%*
In Dexi 1,2,3,4	61.0 G	46.7 M	0.949	+7.6%
In Dexi 1,2,3,4,5	67.1 G	52.4 M	**0.950**	+7.6%
In Dexi 1,2,3,4,5,6	71.5 G	60.4 M	**0.951**	**+7.3%**
In Dexi,2,3,4	60.3 G	45.6 M	0.944	+8.4%
In Dexi 3,4,5	62.8 G	50.3 M	0.941	+9.0%
Location of Non-local Operation:				
After Dexi 2,3 (our setting)	***55.4 G***	***44.5 M***	***0.948***	*+7.8%*
After Dexi 1,2,3	55.6 G	44.6 M	**0.948**	**+7.7%**
After Dexi 2,3,4	56.0 G	45.1 M	0.946	+7.9%
After Dexi 3,4	55.5 G	45.0 M	0.942	+8.5%
After Dexi 4,5	56.1 G	45.4 M	0.939	+8.9%

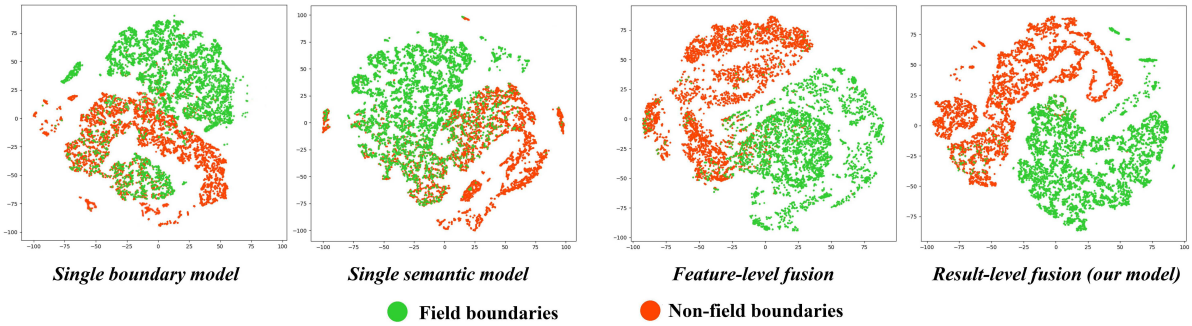

Fig. 17. Visualized feature separability of different models (single boundary and semantic models, the multitask model with feature- and result-level fusion methods).

Furthermore, this visualization also confirms the superiority of our result-level fusion strategy. By using the final, high-confidence region mask to constrain the boundary output—rather than fusing intermediate, less refined features—our model achieves a near-orthogonal representation of the two classes in the feature space. This demonstrates that explicit, high-level semantic gating is a highly effective mechanism for producing accurate and reliable boundary delineation.

B. Architecture Versus Data Scale: A Deeper Look at Occlusion

A provocative finding from our ablation study was the diminished impact of our specialized CRD and NLO modules on the large-scale FTW dataset. This observation prompted a deeper investigation in applied deep learning: are complex challenges like boundary occlusion best solved by innovative architectures or by the scale of data?

Recent landmark studies in large-scale remote sensing analysis suggest that with sufficient data, relatively standard network architectures can indeed achieve remarkable robustness. For instance, Li et al. [49], in their global-scale mapping of 1.4 million lakes, demonstrated that a U-Net model trained on over two decades of satellite data could effectively handle vast complexities. Similarly, studies have shown that deep learning models trained on extensive temporal data can successfully reconstruct significant data gaps (e.g., orbital gaps and cloud cover) without highly specialized modules [30], [31]. These works underscore a critical principle: in a data-rich regime, a network's capacity to learn complex spatiotemporal correlations can often mitigate the need for task-specific modules. Our findings on the FTW dataset align with this, as the performance gap between our full HCFNet and its ablated versions narrows when a large dataset is available, indicating that the network can implicitly learn these image challenges.

However, the success of these large-scale applications hinges on a critical precondition: the availability of vast, comprehensive, and representative training data. Currently, publicly available parcel datasets are mainly concentrated in Europe and other regions characterized by regular, homogenous agricultural landscapes, making a "one-model-fits-all" approach challenging. This precondition of data abundance and landscape diversity is often unmet in high-precision, localized agricultural applications, particularly in regions with fragmented and heterogeneous landscapes like China, where public datasets are scarce. Consequently, a common application paradigm involves training a specialized model on a locally collected, small-scale dataset. It is precisely within this data-scarce regime that the role of architectural inductive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

TABLE XI

BOUNDARY F1-SCORE PERFORMANCE OF HCFNET AND ITS ABLATED VARIANTS UNDER DIFFERENT DATA AVAILABILITY SCENARIOS (SP, DE, AND BE REPRESENT SPAIN, DENMARK, AND BELGIUM, RESPECTIVELY. AVG IS THE AVERAGE SCORE ACROSS THE THREE COUNTRIES)

Train strategies	HCFNet				w/o NLO				w/o CRD			
	Sp	De	Be	Avg	Sp	De	Be	Avg	Sp	De	Be	Avg
U-80	0.933	0.957	0.946	0.945	0.932	0.953	0.947	0.944	0.935	0.955	0.942	0.944
U-50	0.927	0.954	0.940	0.940	0.924	0.951	0.934	0.936	0.920	0.952	0.933	0.935
C-50	0.929	0.955	0.939	0.941	0.915	0.949	0.927	0.930	0.908	0.945	0.923	0.925
C-10	0.919	0.946	0.930	0.932	0.894	0.926	0.904	0.908	0.889	0.923	0.901	0.904
(U-80)-(C-10)	0.014	0.011	0.016	**0.013**	0.038	0.027	0.043	**0.036**	0.046	0.032	0.041	**0.040**

biases, such as those provided by our CRD and NLO modules, becomes indispensable.

To empirically test this data-dependency hypothesis, we designed a targeted experiment. We collected high-resolution (~1 m) Google Earth imagery (occlusion phenomena are more pronounced) and corresponding government-published parcel data in Spain, Denmark, and Belgium. A fixed 30% of the data from each country was held out for validation, while the remaining 70% was pooled to form a training set. From this pool, we simulated both data-rich and data-scarce scenarios.

1) *Unified Models (Data Rich):* A single model was trained on the pooled dataset from all three countries, using either 80% (U-80) or 50% (U-50) of the total training samples;
2) *Country-Specific Models (Data Scarce):* Independent models were trained for each country using only its own data, simulating a localized application. We used 50% (C-50) and 10% (C-10) of each country's training data;

For each of these six strategies, three model variants were trained: the full HCFNet, HCFNet without NLO (w/o NLO), and HCFNet without CRD (w/o CRD). The performance was evaluated using the boundary $F1$-score. As summarized in Table XI, the results offer compelling insights.

In the data-rich (U-80) scenario, the performances of our full HCFNet and its ablated versions converge to nearly identical $F1$ scores. This confirms that with sufficient data, the backbone network learns to handle occlusions and other complexities implicitly, making the specialized modules redundant.

In contrast, as the data become more limited and less diverse (country-specific models), a stark performance gap emerges. This is most evident in the C-10 strategy, representing a realistic data-scarce scenario. Here, the full HCFNet ($F1$ of 0.932) substantially outperforms the models lacking the NLO (0.908) and CRD (0.904) modules. This robustness can be further quantified by the performance degradation from the U-80 to the C-10 setting. The $F1$-score drop for the full HCFNet is merely 0.013, far smaller than the drops for the models without NLO (0.036) and CRD (0.040). This clearly demonstrates that the ablated models fail to generalize, whereas the full HCFNet maintains its robustness, guided by its specialized modules.

In conclusion, while handling occlusion is not a theoretical limitation for deep learning given sufficient data, it remains a critical practical challenge in the majority of real-world applications where data are limited. The CRD and NLO modules were designed not to solve a problem that deep learning cannot theoretically handle but to provide a robust and practical solution for the common scenario of data scarcity. They instill a valuable inductive bias for boundary continuity and long-range feature correlation, effectively compensating for the lack of information in limited training samples.

C. Architectural Rationale of the CRD Module

To validate the specific design choices within the CRD, we performed a more in-depth ablation study in our high-resolution dataset. While it is known that dilated convolutions can improve boundary detection, our CRD module, in contrast to simply stacking standard dilated convolutions, introduces two refinements tailored for field boundaries. First is the evolution of standard dilated convolutions into dilated strip convolutions within our SCP, designed to better capture field edges. Second is the addition of an ADP with stochastically generated voids to explicitly model and reconstruct occluded boundary segments.

The results of this fine-grained ablation, presented in Table XII, validate this design philosophy. We define the network without the SCP and ADP modules as the baseline. While replacing the CRD module with a standard stacked dilation convolution module (SDCM) improves boundary connectivity over the baseline, our SCP with dilated strip convolutions provides a further, significant accuracy gain, confirming its superior adaptation to boundary features. Furthermore, the inclusion of the ADP module with its adaptive void modeling contributes an additional performance boost, demonstrating its effectiveness in explicitly addressing occlusion. This strategy of adaptively modeling the convolution kernel echoes the philosophy of our LCDConv module, where explicitly tailoring the kernel's structure enhances the recognition of specific target features.

D. Additional Attention-Based Modules

In addition to the specialized modules we designed for our network, we also investigated whether other generic attention-based modules could further enhance performance. We tested several common mechanisms that optimize encoded features—such as channel and spatial attention—by integrating them into the network's encoder. As shown in Table XIII, while these attention modules offered marginal improvements to detection quality, their benefits were limited. These minor gains must be weighed against the significant increase in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

TABLE XII

ABLATION STUDY OF CRD MODULE COMPONENTS ON THE HIGH-RESOLUTION DATASET. WE REPORT THE BOUNDARY F1 SCORE TO EVALUATE THE CONTRIBUTION OF EACH COMPONENT

	Baseline	Baseline+SDCM	Baseline+SCP	*Baseline+SCP+ADP*
Boundary F1 score	0.929	0.944	0.946	***0.948***
GRB	+16.3%	+9.6%	+8.4%	***+7.8%***

TABLE XIII

NETWORK COMPLEXITY (FLOPS), PARAMETER COUNT, AVERAGE BOUNDARY ACCURACY (F1 SCORE), AND AVERAGE GRB NUMBER IN NETWORKS WITH DIFFERENT ADDITIONAL ATTENTION MODULES (G:10^9 AND M: 10^6)

Additional Attention-based Modules	FLOPs	Parameters	Average final boundary accuracy (F1)	Average growth ratio of breakpoint number
Non-attention modules (our model)	***55.4 G***	***44.5 M***	*0.948*	*+7.8%*
Squeeze-and-Excitation (SE) [53]	57.2 G	46.1 M	0.949	+8.0%
Efficient Channel Attention (ECA) [54]	57.4 G	46.4 M	0.951	+7.4%
Convolutional Block Attention Module (CBAM) [55]	58.5 G	47.3 M	0.950	+7.8%
Triplet Attention Module (TAM) [56]	59.7 G	48.8 M	0.952	+7.3%
Simple, Parameter-Free Attention Module (simAM) [57]	55.5 G	44.6 M	0.946	+7.7%
Global Attention Mechanism (GAM) [58]	67.5 G	52.0 M	**0.954**	**+7.0%**

TABLE XIV

CROSS-REGIONAL GENERALIZATION PERFORMANCE OF HCFNET. "LS," "JL," AND "DB" ARE LOCALLY TRAINED BENCHMARKS. THE "HT-" PREFIX DENOTES A MODEL TRAINED ON HUANTAI AND TRANSFERRED TO THE TARGET REGION. "FT" INDICATES FINE-TUNING WITH LOCAL DATA

	LS	HT-LS	JL	HT-JL	DB	HT-DB	HT-DB with FT
Boundary F1 score	0.969	0.960	0.945	0.932	0.886	0.819	0.877
GRB	+7.3%	+8.1%	+10.5%	+11.9%	+13.1%	+24.3%	+14.2%

model complexity and parameter counts, necessitating a careful consideration of this tradeoff in any practical application.

E. Regional Transferability

To assess the model's regional transferability, we conducted a cross-regional experiment. Specifically, a model was trained exclusively on data from Huantai County, which is characterized by plains, large parcels, and a wheat–maize rotation. This model was then directly transferred to three other regions.

Liangshan County (HT-LS): An area with an agricultural landscape highly similar to Huantai.

Julu County (HT-JL): An area with a similar cultivation pattern but distinctly smaller parcels.

Dianbai County (HT-DB): A region with a significantly heterogeneous landscape, dominated by mountainous terrain, small irregular parcels, and rice cultivation.

As shown in Table XIV, the model demonstrates excellent direct transferability to Liangshan County, which has a similar landscape, and even to Julu County, where parcel morphology differs. The accuracy drop in both scenarios is negligible compared to that of a locally trained model, proving HCFNet's ability to generalize robustly across similar agricultural systems, even with variations in parcel size.

More notably, when transferred to the highly dissimilar Dianbai County, the model still maintains respectable performance. Furthermore, when a small number of local samples (5%) were used to fine-tune the last few layers of the transferred model (HT-DB with FT), its accuracy increased significantly, nearly matching that of a model trained entirely on local data.

These findings demonstrate HCFNet's strong generalization capability. The model effectively learns the fundamental and transferable features of APs, enabling stable performance across different regions. This allows for direct application in similar landscapes and requires only minimal, efficient adaptation for deployment in vastly different agricultural systems.

F. Conversion to Vector Parcels and Geometric Evaluation

Beyond pixel-level metrics, a comprehensive evaluation requires assessing the geometric integrity of the final vector parcels, which is critical for most downstream applications. This necessitates a postprocessing workflow to convert the raster outputs into topologically complete polygons. A standard approach involves first converting the raster boundary prediction into a set of vector polygons. These are then semantically filtered using the cropland region mask via an object-level majority vote, yielding the final vector parcels.

The geometric integrity of these parcels can then be quantitatively assessed using object-based metrics such as the global overclassification (GOC), global underclassification (GUC), and global total classification (GTC) errors [50]. This geometric evaluation is particularly insightful when comparing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

TABLE XV
COMPARISON OF GEOMETRIC PARCEL INTEGRITY USING OBJECT-BASED METRICS ON THE HIGH-RESOLUTION DATASET. LOWER VALUES INDICATE BETTER PERFORMANCE

Methods	GOC	GUC	GTC
HCFNet	*0.068*	***0.051***	***0.067***
ResUNet-a	0.121	0.208	0.182
BsiNet	0.078	0.129	0.112
SEANet	0.069	0.117	0.103
HBGNet	**0.067**	0.092	0.089

our HCFNet with recent multitask models like HBGNet, which have different architectural philosophies. HBGNet is primarily designed to use its boundary branch to refine the region segmentation for better parcel-level outputs. In contrast, our HCFNet focuses on producing the highest quality boundaries possible, with the hypothesis that superior boundary connectivity is the most critical factor for accurate final parcels.

As detailed in Table XV, our experimental results in the high-resolution dataset validate this hypothesis. HCFNet achieves a significantly lower (better) GUC score (0.051) compared to all competing multitask methods. The GUC metric is particularly telling, as it quantifies undersegmentation errors that occur when fragmented boundaries cause adjacent parcels to be incorrectly merged. This superior GUC score stems directly from our model's ability to generate highly continuous boundaries, which substantially mitigates undersegmentation errors. This reduction in undersegmentation, in turn, leads to a lower overall geometric error (GTC of 0.067).

This advantage is rooted in the object-level fusion workflow strategically leverages the model's dual outputs. It relies on the high-precision boundary prediction to define the crucial geometric structure of parcels while using the region result primarily for semantic labeling. This approach is tolerant of minor inaccuracies in the region mask generated by our lightweight network, as the final parcel geometry is inherited from the more precise boundary map. This once again underscores the foundational importance of achieving high-quality, connected boundaries for successful AP delineation.

G. On the Synergy of the Multitask Framework

A central challenge in designing our multitask model was optimizing the synergy between its components. To this end, we have conducted experiments to determine the most effective combination of our region and boundary branches. This investigation revealed a crucial insight into the relationship between the two tasks: the benefit of the semantic region constraint is not infinite.

Our analysis showed that once the accuracy of the region segmentation task reached an F1 score of approximately 0.86, further improvements in region prediction offered no additional benefit to the final boundary delineation accuracy. This suggests that this accuracy level represents a "good enough" threshold, providing the necessary semantic context to effectively guide the boundary network.

This finding has two strategic implications for future research and model design.

Optimize the Region Branch for Efficiency: Since cropland recognition is a relatively robust task where a simple model can achieve high accuracy [13], [15], the region branch can be further optimized for efficiency in certain application scenarios. For instance, A more lightweight network can be used, or the branch could be eliminated entirely by leveraging reliable, off-the-shelf land cover products as the source of the semantic constraint.

Focus the Boundary Branch on Detail: With the semantic guidance task solved, future research efforts to advance boundary delineation accuracy should shift focus. Instead of pursuing further semantic refinement, the priority should be on enhancing the boundary network's intrinsic ability to perceive and render fine-grained geometric details.

H. Applicability to Medium-Resolution Operational Satellites

The transferability of HCFNet to widely available, medium-resolution operational satellite data is crucial for assessing its broader applicability. We evaluated this by benchmarking the model on the FTW dataset, an extensive collection of 10-m resolution Sentinel-2 imagery covering diverse landscapes. The results (Table I) confirm the model's robustness at this scale, achieving an $F1$ score of 0.899 and a GRB of + 9.4% with bi-temporal inputs.

This performance indicates that the core principles of HCFNet—semantic guidance and connectivity-focused modules—remain effective at coarser resolutions. The primary limitation observed is the inherent difficulty of 10-m resolution data in resolving fine intraparcel boundaries, a factor that can cause a data-label mismatch when ground truth is derived from high-resolution sources. Nevertheless, the model generates topologically coherent parcel maps that accurately represent the landscape patterns visible at this scale. This demonstrates its potential as a cost-effective solution for large-area agricultural monitoring with operational satellite systems.

I. Discussion on Model Limitations and Future Perspectives

While HCFNet has demonstrated robust performance, it is important to acknowledge its limitations and outline avenues for future research. A key consideration is the model's performance under diverse management conditions. The datasets in this study, while varied, primarily feature well-managed agricultural areas with relatively distinct boundaries. In practice, challenges such as mosaic fields, poorly defined boundaries in subsistence farming areas, or boundaries obscured by rampant weed growth can significantly degrade delineation performance. Addressing these scenarios represents a crucial direction for future work. Several promising strategies could be explored. A data-driven approach, involving the collection and annotation of imagery from these challenging regions, would be essential for improving its robustness. Furthermore, architectural enhancements could be investigated; for instance, incorporating modules more sensitive to subtle textural variations, rather than just distinct linear features, might aid in identifying poorly defined boundaries. In addition, super-resolution techniques can be applied as a preprocessing step to enhance the spatial resolution and clarity of the imagery.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

By doing so, indistinct or blurred boundaries are transformed into sharper and more explicit linear features, making them significantly more amenable to accurate delineation.

Another practical challenge is the methodology's reliance on high-resolution optical imagery, which is inherently limited by cloud cover. This poses a constraint for applying the model in tropical and subtropical regions with persistent cloudiness. To enhance the model's all-weather applicability, the integration of synthetic aperture radar (SAR) data, such as from Sentinel-1, is a highly promising solution. SAR's ability to penetrate clouds provides a valuable complementary data source. Future research could explore various SAR–optical fusion strategies within the HCFNet framework. For example, an early fusion approach could involve stacking SAR backscatter data as an additional input channel. Alternatively, a feature-level fusion strategy might prove more effective, where features extracted from both optical and SAR data are intelligently merged within the network's encoder. Overcoming the challenges associated with this integration, such as handling speckle noise and coregistering data from different sensor geometries, would broaden the operational scope of our proposed method.

Finally, while our model produces highly continuous boundaries, future work could also explore network-independent topological postprocessing methods, such as graph-based directional reconnection, to resolve the few remaining minor breaks and guarantee perfect topological integrity for downstream applications.

VI. Conclusion

This study introduces the high-connectivity and accuracy field boundary network (HCFNet) to address the critical challenge of delineating agricultural field boundaries from remote sensing imagery. HCFNet is designed to overcome the limitations of existing methods in balancing semantic accuracy with the preservation of fine, connective boundary details, particularly in the presence of complex linear structures.

HCFNet is built on a two-stage framework that first identifies cropland regions to semantically guide a dedicated boundary delineation task. This boundary task is powered by an enhanced edge network incorporating four key innovations: 1) a LCDConv to precisely capture the linear morphology of boundaries; 2) an NLO to resolve local ambiguities by referencing global context; 3) a CRD to expand the receptive field and explicitly model gaps caused by occlusions; and 4) a HAFM to adaptively fuse multiscale boundary maps for robust performance.

Extensive experiments on diverse datasets, from collected high-resolution imagery to global-scale dataset, confirmed the superiority of HCFNet. It consistently outperformed state-of-the-art models in both boundary accuracy and connectivity, producing significantly less fragmented field networks. Ablation studies validated the efficacy of each component and confirmed that our result-level fusion is more effective than feature-level fusion. A key finding is that while the impact of specialized modules like NLO and CRD may lessen as data availability increases, they are crucial for model robustness in data-scarce scenarios, which is more common in practical applications.

Future work can be pursued in three directions. First, enhancing model robustness in scenarios with ill-defined boundaries via targeted data collection, architectural refinement, and super-resolution preprocessing. Second, exploring SAR–optical data fusion to achieve all-weather applicability. Finally, implementing network-independent topological postprocessing to guarantee perfectly closed polygons for downstream applications.

References

[1] C. Persello, V. A. Tolpekin, J. R. Bergado, and R. A. De By, "Delineation of agricultural fields in smallholder farms from satellite images using fully convolutional networks and combinatorial grouping," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111253, doi: 10.1016/j.rse.2019.111253.

[2] L. Yan and D. P. Roy, "Automated crop field extraction from multi-temporal Web enabled Landsat data," *Remote Sens. Environ.*, vol. 144, pp. 42–64, Mar. 2014, doi: 10.1016/j.rse.2014.01.006.

[3] C. Robb, A. Hardy, J. H. Doonan, and J. Brook, "Semi-automated field plot segmentation from UAS imagery for experimental agriculture," *Frontiers Plant Sci.*, vol. 11, pp. 1–13, Dec. 2020, doi: 10.3389/fpls.2020.591886.

[4] S. Marvaniya, U. Devi, J. Hazra, S. Mujumdar, and N. Gupta, "Small, sparse, but substantial: Techniques for segmenting small agricultural fields using sparse ground data," *Int. J. Remote Sens.*, vol. 42, no. 4, pp. 1512–1534, Feb. 2021, doi: 10.1080/01431161.2020.1834166.

[5] R. Hong, J. Park, S. Jang, H. Shin, H. Kim, and I. Song, "Development of a parcel-level land boundary extraction algorithm for aerial imagery of regularly arranged agricultural areas," *Remote Sens.*, vol. 13, no. 6, p. 1167, Mar. 2021, doi: 10.3390/rs13061167.

[6] Z. Cai et al., "An adaptive image segmentation method with automatic selection of optimal scale for extracting cropland parcels in smallholder farming systems," *Remote Sens.*, vol. 14, no. 13, p. 3067, Jun. 2022, doi: 10.3390/rs14133067.

[7] A. García-Pedrero, C. Gonzalo-Martín, and M. Lillo-Saavedra, "A machine learning approach for agricultural parcel delineation through agglomerative segmentation," *Int. J. Remote Sens.*, vol. 38, no. 7, pp. 1809–1819, Apr. 2017, doi: 10.1080/01431161.2016.1278312.

[8] D. Ming, X. Zhang, M. Wang, and W. Zhou, "Cropland extraction based on OBIA and adaptive scale pre-estimation," *Photogramm. Eng. Remote Sens.*, vol. 82, no. 8, pp. 635–644, Aug. 2016, doi: 10.14358/pers.82.8.635.

[9] M. Mueller, K. Segl, and H. Kaufmann, "Edge- and region-based segmentation technique for the extraction of large, man-made objects in high-resolution satellite imagery," *Pattern Recognit.*, vol. 37, no. 8, pp. 1619–1628, Aug. 2004, doi: 10.1016/j.patcog.2004.03.001.

[10] B. Watkins and A. Van Niekerk, "Automating field boundary delineation with multi-temporal Sentinel-2 imagery," *Comput. Electron. Agricult.*, vol. 167, Dec. 2019, Art. no. 105078, doi: 10.1016/j.compag.2019.105078.

[11] B. Watkins and A. van Niekerk, "A comparison of object-based image analysis approaches for field boundary delineation using multi-temporal Sentinel-2 imagery," *Comput. Electron. Agricult.*, vol. 158, pp. 294–302, Mar. 2019, doi: 10.1016/j.compag.2019.02.009.

[12] Y. Zhu, Y. Pan, T. Hu, D. Zhang, and J. Dai, "An integrated sample-free method for agricultural field delineation from high-resolution remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 13112–13134, 2025, doi: 10.1109/JSTARS.2025.3570750.

[13] D. Zhang et al., "A generalized approach based on convolutional neural networks for large area cropland mapping at very high resolution," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111912, doi: 10.1016/j.rse.2020.111912.

[14] A. Taravat, M. P. Wagner, R. Bonifacio, and D. Petit, "Advanced fully convolutional networks for agricultural field boundary detection," *Remote Sens.*, vol. 13, no. 4, p. 722, Feb. 2021, doi: 10.3390/rs13040722.

[15] W. Zhang et al., "A novel knowledge-driven automated solution for high-resolution cropland extraction by cross-scale sample transfer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2024, doi: 10.1109/TGRS.2023.3299956.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

[16] L. Xia, J. Luo, Y. Sun, and H. Yang, "Deep extraction of cropland parcels from very high-resolution remotely sensed imagery," in *Proc. 7th Int. Conf. Agro-Geoinformatics (Agro-geoinformatics)*, Aug. 2018, pp. 1–5, doi: 10.1109/Agro-Geoinformatics.2018.8476002.

[17] L. Xu, D. Ming, T. Du, Y. Chen, D. Dong, and C. Zhou, "Delineation of cultivated land parcels based on deep convolutional networks and geographical thematic scene division of remotely sensed images," *Comput. Electron. Agricult.*, vol. 192, Jan. 2022, Art. no. 106611, doi: 10.1016/j.compag.2021.106611.

[18] K. M. Masoud, C. Persello, and V. A. Tolpekin, "Delineation of agricultural field boundaries from Sentinel-2 images using a novel super-resolution contour detector based on fully convolutional networks," *Remote Sens.*, vol. 12, no. 1, p. 59, Dec. 2019, doi: 10.3390/rs12010059.

[19] X. Xia, C. Persello, and M. Koeva, "Deep fully convolutional networks for cadastral boundary detection from UAV images," *Remote Sens.*, vol. 11, no. 14, p. 1725, Jul. 2019, doi: 10.3390/rs11141725.

[20] B. Fetai, M. Račič, and A. Lisec, "Deep learning for detection of visible land boundaries from UAV imagery," *Remote Sens.*, vol. 13, no. 11, p. 2077, May 2021, doi: 10.3390/rs13112077.

[21] D. K. Gopidas et al., "Integrated deep learning based segmentation and classification method for boundary delineation of agricultural fields in multitemporal satellite images," *Int. J. Mod. Agric.*, vol. 10, no. 2, pp. 1804–1822, 2021.

[22] R. Yang, Z. U. Ahmed, U. C. Schulthess, M. Kamal, and R. Rai, "Detecting functional field units from satellite images in smallholder farming systems using a deep learning based computer vision approach: A case study from Bangladesh," *Remote Sens. Appl., Soc. Environ.*, vol. 20, Nov. 2020, Art. no. 100413, doi: 10.1016/j.rsase.2020.100413.

[23] H. Zhang et al., "Automated delineation of agricultural field boundaries from Sentinel-2 images using recurrent residual U-Net," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102557, doi: 10.1016/j.jag.2021.102557.

[24] F. Waldner et al., "Detect, consolidate, delineate: Scalable mapping of field boundaries using satellite images," *Remote Sens.*, vol. 13, no. 11, p. 2197, Jun. 2021, doi: 10.3390/rs13112197.

[25] J. Long, M. Li, X. Wang, and A. Stein, "Delineation of agricultural fields using multi-task BsiNet from high-resolution satellite images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102871, doi: 10.1016/j.jag.2022.102871.

[26] M. Li, J. Long, A. Stein, and X. Wang, "Using a semantic edge-aware multi-task neural network to delineate agricultural parcels from remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 200, pp. 24–40, Jun. 2023, doi: 10.1016/j.isprsjprs.2023.04.019.

[27] J. Long, H. Zhao, M. Li, X. Wang, and C. Lu, "Integrating segment anything model derived boundary prior and high-level semantics for cropland extraction from high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: 10.1109/LGRS.2024.3454263.

[28] H. Zhao et al., "A large-scale VHR parcel dataset and a novel hierarchical semantic boundary-guided network for agricultural parcel delineation," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 1–19, Mar. 2025, doi: 10.1016/j.isprsjprs.2025.01.034.

[29] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[30] H. Wang, Y. Zhou, and X. Li, "GDCM: Generalized data completion model for satellite observations," *Remote Sens. Environ.*, vol. 324, Jul. 2025, Art. no. 114760, doi: 10.1016/j.rse.2025.114760.

[31] E. Jang, D. Han, J. Im, T. Sung, and Y. Jun Kim, "Deep learning-based gap filling for near real-time seamless daily global sea surface salinity using satellite observations," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104029, doi: 10.1016/j.jag.2024.104029.

[32] X. Soria, E. Riba, and A. Sappa, "Dense extreme inception network: Towards a robust CNN model for edge detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 1912–1921, doi: 10.1109/WACV45572.2020.9093290.

[33] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803, doi: 10.1109/CVPR.2018.00813.

[34] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet—A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020, doi: 10.1016/j.isprsjprs.2020.01.013.

[35] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571, doi: 10.1109/3DV.2016.79.

[36] Y. Liu et al., "Richer convolutional features for edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 8, pp. 1939–1946, Aug. 2019, doi: 10.1109/TPAMI.2018.2878849.

[37] A. Kendall, Y. Gal, and R. Cipolla, "Kendall_multi-task_learning_using_CVPR_2018_paper.pdf," in *Proc. CVPR*, Jun. 2018, pp. 7482–7491.

[38] H. Wang, Y. Zhu, H. Adam, A. Yuille, and L.-C. Chen, "MaX-DeepLab: End-to-end panoptic segmentation with mask transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5459–5470, doi: 10.1109/CVPR46437.2021.00542.

[39] B. Cheng et al., "Panoptic-DeepLab: A simple, strong, and fast baseline for bottom-up panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12472–12482, doi: 10.1109/CVPR42600.2020.01249.

[40] J. Long, S. Liu, M. Li, H. Zhao, and Y. Jin, "BGSNet: A boundary-guided Siamese multitask network for semantic change detection from high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 221–237, Jul. 2025, doi: 10.1016/j.isprsjprs.2025.04.030.

[41] Y. Zhu, Y. Pan, D. Zhang, H. Wu, and C. Zhao, "A deep learning method for cultivated land parcels' (CLPs) delineation from high-resolution remote sensing images with high-generalization capability," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–25, 2024, doi: 10.1109/TGRS.2024.3425673.

[42] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent.*, 2015, pp. 1–15.

[43] IAAO. (Jan. 2015). *Standard on Digital Cadastral Maps and Parcel Identifiers*. [Online]. Available: https://www.iaao.org/media/standards/Standard_Digital_Cadastral_Maps_2015.pdf

[44] Y. Zhu, Y. Pan, T. Hu, D. Zhang, C. Zhao, and Y. Gao, "A generalized framework for agricultural field delineation from high-resolution satellite imageries," *Int. J. Digit. Earth*, vol. 17, no. 1, pp. 1–31, Dec. 2024, doi: 10.1080/17538947.2023.2297947.

[45] F. Waldner and F. I. Diakogiannis, "Deep learning on edge: Extracting field boundaries from satellite images with a convolutional neural network," *Remote Sens. Environ.*, vol. 245, Aug. 2020, Art. no. 111741, doi: 10.1016/j.rse.2020.111741.

[46] S. Liu, L. Liu, F. Xu, J. Chen, Y. Yuan, and X. Chen, "A deep learning method for individual arable field (IAF) extraction with cross-domain adversarial capability," *Comput. Electron. Agricult.*, vol. 203, Dec. 2022, Art. no. 107473, doi: 10.1016/j.compag.2022.107473.

[47] A. O. Onojeghuo, Y. Miao, and G. A. Blackburn, "Deep ResU-Net convolutional neural networks segmentation for smallholder paddy Rice mapping using sentinel 1 SAR and sentinel 2 optical imagery," *Remote Sens.*, vol. 15, no. 6, pp. 1–20, Mar. 2023, doi: 10.3390/rs15061517.

[48] L. V. D. Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008.

[49] L. Li, D. Long, Y. Wang, and R. I. Woolway, "Global dominance of seasonality in shaping lake-surface-extent dynamics," *Nature*, vol. 642, no. 8067, pp. 361–368, Jun. 2025, doi: 10.1038/s41586-025-09046-3.

[50] C. Persello and L. Bruzzone, "A novel protocol for accuracy assessment in classification of very high resolution images," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 3, pp. 1232–1244, Mar. 2010, doi: 10.1109/TGRS.2009.2029570.

[51] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141. [Online]. Available: http://openaccess.thecvf.com/content_cvpr_2018/html/Hu_Squeeze-and-Excitation_Networks_CVPR_2018_paper.html

[52] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11531–11539, doi: 10.1109/CVPR42600.2020.01155.

[53] S. Woo, J. Park, J. Lee, and I. S. Kweon, "Convolutional_block_attention," in *Proc. ECCV*, 2018, pp. 1–17. [Online]. Available: http://files/737/Wooetal.-ConvolutionalBlockAttentionModule.pdf

[54] D. Misra, T. Nalamada, A. U. Arasanipalai, and Q. Hou, "Rotate to attend: Convolutional triplet attention module," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Waikoloa, HI, USA, Jan. 2021, pp. 3138–3147, doi: 10.1109/WACV48630.2021.00318.

[55] L. Yang, R. Zhang, L. Li, and X. Xie, "SimAM: A simple, parameter-free attention module for convolutional neural networks," in *Proc. 38th Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 11863–11874.

[56] Y. Liu, Z. Shao, and N. Hoffmann, "Global attention mechanism: Retain information to enhance channel-spatial interactions," 2021, *arXiv:2112.05561*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.

Yu Zhu received the B.S. degree in geography from China University of Geosciences, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree in remote sensing science and technology with the State Key Laboratory of Remote Sensing Science, Faculty of Geographical Science, Beijing Normal University, Beijing.

His research interests include deep learning-based image processing and agricultural intelligent remote sensing.

Yaozhong Pan received the B.Sc. degree in remote sensing technology from Zhejiang University, Hangzhou, China, in 1988, and the M.S. and Ph.D. degrees in physical geography from Beijing Normal University, Beijing, China, in 1994 and 1997, respectively.

He is currently a Professor with Beijing Normal University, where he is currently the Dean of the Institute of Remote Sensing Science and Engineering. He has authored or co-authored more than 150 peer-reviewed international journal articles. His research interests include agricultural intelligent remote sensing and real-time remote sensing of surface anomalies.

Dr. Pan has won several research awards, including the Precursor of the Science and Technology Production Transformation in Beijing, awarded by Beijing Economy Committee, Beijing Science and Technology Committee, and Beijing Education Committee in 2002, the First Prize of National Natural Science Award superscripted by the Ministry of Education in 2004, and the 8th National Youth Geographic Science and Technology Award issued by the Ministry of Education in 2005.

Tangao Hu (Member, IEEE) received the B.S. degree in environmental engineering from China University of Mining and Technology, Beijing, China, in 2006, and the Ph.D. degree in geoscience and remote sensing from Beijing Normal University, Beijing, in 2011.

He is currently a Professor at the Institute of Remote Sensing and Earth Sciences, Hangzhou Normal University, Hangzhou, China. He has authored or co-authored more than 50 peer-reviewed international journal articles. His main research interests are urban flood monitoring and land-cover and land-use change detection.

Yao Liu received the B.S. degree in remote sensing science and technology from Shandong University of Science and Technology, Qingdao, China, in 2020. She is currently pursuing the Ph.D. degree in remote sensing science and technology with the State Key Laboratory of Remote Sensing Science, Faculty of Geographical Science, Beijing Normal University, Beijing, China.

Her research interests include hydrological modeling, deep learning-based image processing, and information extraction.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:32:31 UTC from IEEE Xplore. Restrictions apply.